Drive various types of service vehicles

D2.TTG.CL3.09

Trainee Manual





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William Angliss Institute
Specialist centre for foods, tourism & hospitality

Drive various types of service vehicles

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Trainee Manual

Project Base

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The Association of Southeast Asian Nations (ASEAN) was established on 8 August 1967. The Member States of the Association are Brunei Darussalam, Cambodia, Indonesia, Lao PDR, Malaysia, Myanmar, Philippines, Singapore, Thailand and Viet Nam.

The ASEAN Secretariat is based in Jakarta, Indonesia.

General Information on ASEAN appears online at the ASEAN Website: www.asean.org.

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Introduction to trainee manual

To the Trainee

Congratulations on joining this course. This Trainee Manual is one part of a 'toolbox' which is a resource provided to trainees, trainers and assessors to help you become competent in various areas of your work.

The 'toolbox' consists of three elements:

- A Trainee Manual for you to read and study at home or in class
- A Trainer Guide with Power Point slides to help your Trainer explain the content of the training material and provide class activities to help with practice
- An Assessment Manual which provides your Assessor with oral and written questions and other assessment tasks to establish whether or not you have achieved competency.

The first thing you may notice is that this training program and the information you find in the Trainee Manual seems different to the textbooks you have used previously. This is because the method of instruction and examination is different. The method used is called Competency based training (CBT) and Competency based assessment (CBA). CBT and CBA is the training and assessment system chosen by ASEAN (Association of South-East Asian Nations) to train people to work in the tourism and hospitality industry throughout all the ASEAN member states.

What is the CBT and CBA system and why has it been adopted by ASEAN?

CBT is a way of training that concentrates on what a worker can do or is required to do at work. The aim is of the training is to enable trainees to perform tasks and duties at a standard expected by employers. CBT seeks to develop the skills, knowledge and attitudes (or recognise the ones the trainee already possesses) to achieve the required competency standard. ASEAN has adopted the CBT/CBA training system as it is able to produce the type of worker that industry is looking for and this therefore increases trainee chances of obtaining employment.

CBA involves collecting evidence and making a judgement of the extent to which a worker can perform his/her duties at the required competency standard. Where a trainee can already demonstrate a degree of competency, either due to prior training or work experience, a process of 'Recognition of Prior Learning' (RPL) is available to trainees to recognise this. Please speak to your trainer about RPL if you think this applies to you.

What is a competency standard?

Competency standards are descriptions of the skills and knowledge required to perform a task or activity at the level of a required standard.

242 competency standards for the tourism and hospitality industries throughout the ASEAN region have been developed to cover all the knowledge, skills and attitudes required to work in the following occupational areas:

- Housekeeping
- Food Production
- Food and Beverage Service
- Front Office

- Travel Agencies
- Tour Operations.

All of these competency standards are available for you to look at. In fact you will find a summary of each one at the beginning of each Trainee Manual under the heading ‘Unit Descriptor’. The unit descriptor describes the content of the unit you will be studying in the Trainee Manual and provides a table of contents which are divided up into ‘Elements’ and ‘Performance Criteria”. An element is a description of one aspect of what has to be achieved in the workplace. The ‘Performance Criteria’ below each element details the level of performance that needs to be demonstrated to be declared competent.

There are other components of the competency standard:

- *Unit Title:* statement about what is to be done in the workplace
- *Unit Number:* unique number identifying the particular competency
- *Nominal hours:* number of classroom or practical hours usually needed to complete the competency. We call them ‘nominal’ hours because they can vary e.g. sometimes it will take an individual less time to complete a unit of competency because he/she has prior knowledge or work experience in that area.

The final heading you will see before you start reading the Trainee Manual is the ‘Assessment Matrix’. Competency based assessment requires trainees to be assessed in at least 2 – 3 different ways, one of which must be practical. This section outlines three ways assessment can be carried out and includes work projects, written questions and oral questions. The matrix is designed to show you which performance criteria will be assessed and how they will be assessed. Your trainer and/or assessor may also use other assessment methods including ‘Observation Checklist’ and ‘Third Party Statement’. An observation checklist is a way of recording how you perform at work and a third party statement is a statement by a supervisor or employer about the degree of competence they believe you have achieved. This can be based on observing your workplace performance, inspecting your work or gaining feedback from fellow workers.

Your trainer and/or assessor may use other methods to assess you such as:

- Journals
- Oral presentations
- Role plays
- Log books
- Group projects
- Practical demonstrations.

Remember your trainer is there to help you succeed and become competent. Please feel free to ask him or her for more explanation of what you have just read and of what is expected from you and best wishes for your future studies and future career in tourism and hospitality.

Unit descriptor

Drive various types of service vehicles

This unit deals with the skills and knowledge required to Drive various types of service vehicles in a range of settings within the travel industries workplace context.

Unit Code:

D2.TTG.CL3.09

Nominal Hours:

120

Element 1: Obtain driver's licence(s)

Performance Criteria

1.1 Identify the vehicles that have to be driven

1.2 Identify the driver's licence(s) that need to be obtained

1.3 Undertake training to obtain the necessary licence(s)

1.4 Undertake driver licence assessment successfully

Element 2: Monitor service vehicles

Performance Criteria

2.1 Perform regular preventative maintenance and operational checks of designated small vehicles

2.2 Complete and retain service reports as required

2.3 Adhere to scheduled maintenance and service requirements as set out by the vehicle manufacturer

2.4 Conduct special checks and service after designated trips

2.5 Check performance and efficiency of vehicle during operation

2.6 Undertake minor routine repairs

Element 3: Monitor road conditions

Performance Criteria

3.1 Assess road conditions

3.2 Assess traffic conditions

3.3 Feedback information to head office about road and traffic conditions

3.4 Identify factors that may cause delays or route deviations

3.5 Factor conditions into choice of route to be taken

3.6 Read a map

Element 4: Drive vehicles

Performance Criteria

4.1 Demonstrate basic operational driving practices

4.2 Demonstrate practices related to the towing of a trailer

4.3 Demonstrate reversing practices

4.4 Demonstrate parking and shutting down procedures

4.5 Identify and modify driving techniques to accommodate driving hazards

4.6 Modify driving style to accommodate identified vehicle problems and faults

4.7 Follow emergency procedures

Assessment matrix

Showing mapping of Performance Criteria against Work Projects, Written Questions and Oral Questions

The Assessment Matrix indicates three of the most common assessment activities your Assessor may use to assess your understanding of the content of this manual and your performance - Work Projects, Written Questions and Oral Questions. It also indicates where you can find the subject content related to these assessment activities in the Trainee Manual (i.e. under which element or performance criteria). As explained in the Introduction, however, the assessors are free to choose which assessment activities are most suitable to best capture evidence of competency as they deem appropriate for individual students.

		Work Projects	Written Questions	Oral Questions
Element 1: Obtain driver's licence(s)				
1.1	Identify the vehicles that have to be driven	1.1	1	1,2
1.2	Identify the driver's licence(s) that need to be obtained	1.1	2	3,4
1.3	Undertake training to obtain the necessary licence(s)	1.1	3	5,6
1.4	Undertake driver licence assessment successfully	1.1	4	7,8
Element 2: Monitor service vehicles				
2.1	Perform regular preventative maintenance and operational checks of designated small vehicles	2.1	5	9,10,11,12
2.2	Complete and retain service reports as required	2.1	6	13,14
2.3	Adhere to scheduled maintenance and service requirements as set out by the vehicle manufacturer	2.1	7	15,16
2.4	Conduct special checks and service after designated trips	2.1	8	17,18
2.5	Check performance and efficiency of vehicle during operation	2.1	9	19,20
2.6	Undertake minor routine repairs	2.1	10	21,22

		Work Projects	Written Questions	Oral Questions
Element 3: Monitor road conditions				
3.1	Assess road conditions	3.1	11	23,24
3.2	Assess traffic conditions	3.1	12	25,26
3.3	Feedback information to head office about road and traffic conditions	3.1	13	27,28
3.4	Identify factors that may cause delays or route deviations	3.1	14	29,30
3.5	Factor conditions into choice of route to be taken	3.1	15	31,32
3.6	Read a map	3.2	16	33,34
Element 4: Drive vehicles				
4.1	Demonstrate basic operational driving practices	4.1	17	35,36
4.2	Demonstrate practices related to the towing of a trailer	4.1	18	37,38
4.3	Demonstrate reversing practices	4.1	19	39,40
4.4	Demonstrate parking and shutting down procedures	4.1	20	41,42
4.5	Identify and modify driving techniques to accommodate driving hazards	4.1	21	43,44
4.6	Modify driving style to accommodate identified vehicle problems and faults	4.1	22	45,46
4.7	Follow emergency procedures	4.1	23,24	47,48

Glossary

Term	Explanation
4WD	Four-wheel drive; 4-wheel drive
AWD	All-wheel drive
A4WD	Automatic 4-wheel drive
CAS	Crash Avoidance Space
CB	Citizens' Band
Duty of Care	A legally imposed obligation on the business and staff to take reasonable care to avoid causing foreseeable harm to customers/people.
HT	High Tension
'L' plates	Learner (driver) plates
OBD	On Board Diagnostics
Pax	Passengers
PDL	Provisional Driving Licence
SOP	Standard Operating Procedure
UHF	Ultra High Frequency

Element 1:
Obtain driver's licence(s)

1.1 Identify the vehicles that have to be driven

Introduction

In order to obtain relevant driving licences appropriate to the individual place of employment it is necessary to identify the vehicles which may need to be driven.

This section defines service vehicles, explains the importance of identifying service vehicles which may have to be driven, presents a list of possible vehicles and discusses ways to identify those vehicles.

Service vehicles defined

Service vehicles may be seen as:

- Any vehicle the tour operator/travel agency uses to conduct its day-to-day business – such as:
  - Visiting potential clients and other businesses
  - Undertaking general everyday tasks including banking, post, obtaining supplies
  - Scouting locations, sites and destinations
- Any vehicle driven by tour guides while on tour
- Any vehicle used to transfer passengers – for example, from hotels to a tour mustering point
- Any vehicle that provides support to a tour – such as transportation of gear and/or catering.

Importance of identifying service vehicles to be driven

It is important to identify the service vehicles which may need to be driven in order to:

- Get to know the vehicles which are available
- Determine the licence type/s required for driving vehicles which are part of the job
- Identify *additional* driving licences which should be considered/obtained – in order to:
  - Make current employment more secure
  - Open up new career opportunities
- Obtain driving licences required by the job/employer.

List of possible service vehicles

Different businesses will have different vehicles.

The types and numbers of vehicles they use will be determined by a combination of the following factors:

- The types of tours they offer
- Frequency and duration of tours
- Number of passengers carried
- Destinations visited and condition/type of roads travelled
- Extent to which they use vehicles provided by third party operators.

Tour operators and travel agencies may use:

- Cars – being sedans which may be:
  - Petrol or diesel
  - Front-wheel drive or rear-wheel drive
- Four-wheel drives – which may be:
  - Petrol or diesel
  - Partial 4WD
  - Permanent all-wheel drive
  - Automatic 4-wheel drive
- Utilities – which can be used:
  - To transport relatively small items and/or relatively small amounts of items
  - As general purposes vehicles
- Light and heavy commercial vehicles – such as:
  - Vans
  - Vehicles capable of carrying up to 12 people/pax
  - Vehicles up to a given laden and/or un-laden weight as prescribed by host country legislation.

Ways to identify service vehicles to be driven

To identify the vehicles which need to be driven:

- Speak with business owner/manager
- Talk with supervisor
- Visit the location where vehicles are garaged
- Read materials detailing tours and services provided by the organisation.

1.2 Identify the driver's licence(s) that need to be obtained

Introduction

Across all ASEAN countries there are a variety of classifications of driving licences.

This section provides an overview of the licences available, presents actions which will help determine the required licence and presents information regarding possible licences available.

Overview

It is essential to understand:

- Under a 1985 ASEAN agreement the drivers' licences of Brunei Darussalam, the Republic of Indonesia, Malaysia, the Republic of the Philippines, the Republic of Singapore and the Kingdom of Thailand are recognised by all those ASEAN Member States
- Some ASEAN countries will accept and recognise driver's licences issued from *any* other ASEAN country

- A new ASEAN driving licence issued by Thailand has been approved for use in all 10 ASEAN Member States
- Standard 4WD vehicles do not require a separate licence
- A special licence is required to drive:
  - Buses/coaches or vehicles carrying paying customers/passengers – this means a 'car' licence or a motorcycle' licence is not sufficient/not acceptable
  - Light and heavy vehicles – such as trucks and large vans
- People seeking a driving licence may be required to obtain a Learner's Driving Licence – meaning:
  - They need to pass a Theory Test
  - They can drive a vehicle to gain experience and expertise only under the guidance and supervision of a fully licensed driver
  - This licence is only valid in the country of issue
- When the initial driving licence is granted it may take the form of a Probationary licence – meaning the holder is subject to special obligations for a given period of 1 – 3 years
- Manual and automatic transmission vehicles are treated differently – in that:
  - A person who holds an 'automatic only' licence cannot drive a manual vehicle
  - A person holding a manual licence is allowed to drive a manual vehicle
- A 'full' driving licence will/may only be issued when the requirements of the Probationary licence have been met
- Licences are not 'permanent' – they are issued for a limited period of time (say, 5 years) and must be renewed following host country re-application protocols/renewal procedures

- People of a certain age (say, 70+ years) may be required to obtain a medical certificate stating their fitness to drive
- Driving is a privilege not a right – the law has penalties for failure to comply with host country road legislation, such as:
  - Demerit points – such that when a maximum number of points has been given to the driver for driving offence their licence will be suspended or cancelled
  - Monetary fines for failing to comply with road laws.

Licence classifications

It is possible countries make available driving licences in Classes such as:

- Motor cycle – different licence types for bikes of varying engine capacities
- Private motor cars/vehicles – with possible variations, restrictions/limitations centred on:
  - Vehicle weight when un-laden – for example, 2,500kg
  - Maximum number of vehicle passengers – such as 7 passengers excluding the driver
  - Whether vehicle is a manual or automatic vehicle

- Commercial car – for operation of a taxi
- Buses and coaches
- Light vehicles – being:
  - Vans
  - Lorries under a certain weight
- Medium vehicles – which may include:
  - Non-reticulated lorries below a nominated length and/or weight
- Heavy vehicles – which apply to, for example:
  - The largest trucks
  - Reticulated vehicles.

Websites

It is interesting to visit:

- http://treaty.kemlu.go.id/uploads-pub/5121_ASEAN-1985-0051.pdf - Agreement on the recognition of domestic driving licences issued by ASEAN countries
- http://www.thaivisa.com/forum/topic/833756-new-driver-license-is-valid-in-asean-member-countries/.

Licence identification

To fully identify the specific driving licence requirements for individual countries it is necessary to:

- Contact the relevant Land Transportation Office or similar
- Determine if the country requires:
  - A Public Service Vehicle Licence – for example, see http://driving-school.com.my/driving-license/how-to-secure-a-malaysian-public-service-vehicle-license/
  - A special licence or permit – such as a Bus Driver's Vocational Licence, see http://www.lta.gov.sg/content/ltaweb/en/industry-matters/vocational-licences.html.

Application pre-requisites

Always check with local Authorities when applying for any driving licence as requirements may:

- Change over time
- Vary between countries.

Indicative pre-requisites when applying for a driving licence may be:

- To make application on a nominated form – supplying as/if required:
  - Personal and identity details
  - Driving history information
- To pay a stated fee
- To attend and successfully complete a designated driver training course
- To be of a minimum age – such as 18 or 21 years of age: a Learner's permit/licence may be available from local authority/Road Transport Administration when younger than this (such as 15 or 16 years of age)
- To hold a nominated other Class of licence – such as a car licence for a given period of time (say for one year) if applying for a truck or bus licence
- To have a clean driving record – in terms (for example) of not having had licence suspended
- To provide approved medical report showing fitness to drive.

1.3 Undertake training to obtain the necessary licence(s)

Introduction

The need to complete formal training and assessment is a standard pre-requisite in order to obtain a driving licence.

This section describes context and content for this training.

Context

Training for applicants for a driver's licence can be provided by:

- Private companies operating as licensed Driving schools or Driving Centres
- Licensed individuals operating as private driving instructors.

Dual-control cars may be used.

Training is generally:

- Arranged at a time to suit the individual
- Paid for on a per session basis
- Conducted in the local area
- Tailored to suit the individual needs and experience of the person wishing to obtain the licence
- Organised to cover a range of different driving conditions, traffic patterns and typical, day-to-day eventualities
- Courses are generally conducted at scheduled and regular dates providing there are sufficient enrolments to allow the course to be conducted.

Learner drivers

A Learner's permit/licence (or Provisional Driving Licence [PDL]) is typically required by those who are learning to drive/practising to obtain their driver's licence.

When holding a Learner's permit/licence there **may** be a need for the driver to:

- Display 'L' plates at the front and rear of the car so they are readily visible to other drivers – indicating the driver of the vehicle is not yet fully qualified/competent and should be given some extra consideration
- Be accompanied at all times by a fully qualified driver whose job it is to provide instruction, guidance and direction
- Observe restrictions relating to the number of passengers (and their age) in their vehicle
- Not tow a trailer
- Have zero blood alcohol.

In some countries, Learner drivers may be banned from driving on certain roads such as Expressways where the speed and flow of traffic is deemed too high.

Inform trainees on context Content

Driver training commonly features two separate components:

- Theory
- Practical.

Theory

The theory component may address topics such as:

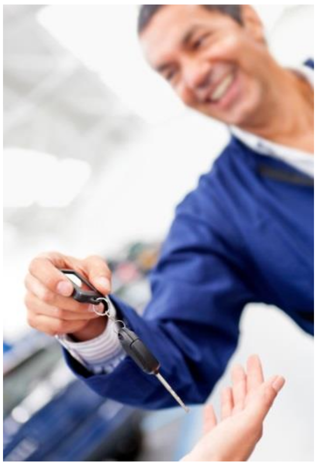
- Basic rules and responsibilities
- Alcohol, drugs, medicine and driving
- Need for and use of seat belts
- Meaning of road signs
- Speed limits
- Give way rules
- Road marking
- Traffic control/traffic lights
- Road courtesy
- Safe and/or defensive driving
- Parking
- Offences and penalties
- Legal responsibilities.

Practical

This section is where trainers demonstrate driving skills and students are given opportunity to practice, for example:

- Pre-drive checks – where the controls of the vehicle are identified, set and/or checked/verified
- Basic level tasks – such as driving in light traffic and:
  - Starting and travelling at an appropriate speed relative to the conditions and the posted speed limit
  - Stopping as and when required maintaining an appropriate distance between vehicles
  - Observing traffic, pedestrians and other road users
  - Turning left and right using indicators as required
  - Changing lanes and staying in the correct lane
  - Reverse parking
  - Executing a three-point turn
  - Demonstrating overall control of vehicle

- Advanced level tasks – such as driving in heavy traffic and:
  - All the above
  - Merging with other traffic
- Using manual and automatic transmission vehicles – as appropriate to the type/class of licence being sought.



It can also be a requirement the Learner attains a minimum of prescribed experience – such as a stated number of hours of experience under approved supervision/instruction (and perhaps this may need to be recorded in a Log Book):

- Of day driving
- Of night driving
- In a variety of conditions – such as:
  - On-road and off-road
  - In wet and dry weather.

1.4 Undertake driver licence assessment successfully

Introduction

There is a need to successfully complete prescribed driving licence for the specific licence type/class which is being sought before the licence will be granted/issued.

This section indicates requirements which may apply.

Possible pre-requisites

In order for Authorities to assess an applicant as suitable and competent to hold a driver's licence they may/will require:

- Properly and fully completed application form – as provided by the Authority which may be paper-based or in electronic format
- Payment of a fee – which can be separated into components for:
  - The assessment
  - The licence itself for a given period of time/number of years

- Evidence of successful completion of a nominated driver training course
- Proof of identity or age
- Evidence of having held a nominated other/lower classification of licence for a stated period – such as a Learner's permit/licence
- Medical report stating that, in the opinion of the designated medical practitioner/examiner, the applicant is fit to hold such a licence
- Copy of passport and/or work or other visa
- Satisfactory completion of a Theory test – where for example:
  - A stated number of questions must be attempted in a given time
  - A nominated 'Pass mark' must be obtained
- Satisfactory completion of a Practical test – where for example:
  - A designated route must be driven observing all applicable road signs and laws
  - Specified manoeuvres must be completed within a given time.

Failure

Where an applicant fails the test to obtain a driving licence, the following usually applies:

- Another fee is payable for re-application
- Another application has to be made
- There may be a minimum waiting time between applications
- If the Theory test was passed but the Practical test was failed, only the Practical test may need to be done
- The Theory test must be passed before the Practical test can be attempted.

Work Projects

It is a requirement of this Unit you complete Work Projects as advised by your Trainer. You must submit documentation, suitable evidence or other relevant proof of completion of the project to your Trainer by the agreed date.

1.1 To fulfil the requirements of this Work Project you are required to obtain a light commercial vehicle driving licence providing evidence you have:

- Listed the vehicles the vehicles which can be driven using the licence
- Named additional driver's licence(s) available for buses/coaches
- Undertaken training to obtain the licence.

Summary

Obtain driver's licence(s)

When obtaining driver's licence(s):

- Speak to management to identify the exact types of vehicles which need to be driven
- Determine the in-country licences necessary to enable legal driving of all vehicles which need to be driven
- Make enquiries to find out the requirements that relate to applying for necessary driving licences such as pre-requisite licences, age restrictions and medical certification
- Determine providers for necessary training
- Undertake required training until successfully completed
- Apply formally for required driving licences including payment of relevant fees
- Complete mandatory theory and practical testing to gain required licences
- Provide evidence of successful completion to employer
- Carry driver's licence at all times when driving vehicles.

Element 2: Monitor service vehicles

2.1 Perform regular preventative maintenance and operational checks of designated small vehicles

Introduction

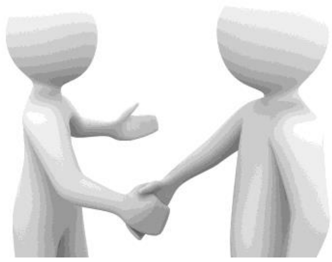
Conducting regular inspections and vehicle checks is a fundamental element in monitoring the condition and operation of service vehicles.

This section examines the practice of conducting visual inspections and describes a range of checks which are commonly made.

Important points

In relation to this topic it is important to understand:

- This unit is not intended to provide the training needed for becoming a fully-qualified/A-grade mechanic. To become a properly trained professional there is a need to undertake a three or four year apprenticeship that covers much more than is contained in this unit and requires many more hours of practical experience
- The manual and the notes cannot address the requirements for all types, makes and models of service vehicles. There are too many choices in the marketplace, too many variations, and too many after-market and up-grade options



- Cleaning is an important element of servicing and repairing vehicles in two ways:
  - Cleaning prior to performing inspections and repairs will make it easier to identify problems and diagnose issues
  - Cleaning of certain components is an essential part of the preventative maintenance process as it removes dirt, dust, grit, sand and other materials which can interfere with and damage correct operation of items
- Inspections/checks of service vehicles may be required to be conducted:
  - At the start of each day a vehicle is used
  - One some other basis – such as every 2nd or 3rd day, or weekly
  - Before nominated trips/tours or other activities such as external hiring to another business/operator
- More detail on some aspects covered by these notes can be found in the unit ‘Carry out vehicle maintenance or minor repairs’.

Doing a visual inspection

The inspection process should/could involve:

- Looking at and under the vehicle and closely checking all aspects of the vehicle – while it is on the ground, and/or over an inspection pit
- Lighting the areas being inspected – with inspection lights or torches
- Using a checklist – to provide evidence the inspection occurred and note defects and matters requiring attention
- Inspecting the vehicle in pairs – so that issues can be analysed/discussed and there is less chance something will be missed
- Looking for signs of damage to the vehicle – as these signs can indicate:
  - Where a more detailed inspection needs to occur to determine:
    - If the damage has caused a problem/further problem
    - Whether the *damage is a result* of another issue
  - Need for immediate action – to:
    - Fix the cause
    - Take temporary action to reduce/prevent further damage
    - Keep the vehicle on-the-road/operational
  - Need to take later action – which may not be *immediately* necessary to
  - Possible cause or source of the damage
  - What may be needed/possible to prevent recurrence
- Checking for missing parts – from:
  - Inside the vehicle
  - On the exterior of the vehicle
  - Under the hood/bonnet
  - Elsewhere on the vehicle
- Noting leaking fluids – as this can:
  - Identify type of fluid
  - Identify location of leak
  - Indicate severity of problem
  - Determine, to an extent, the type of problem needing to be addressed
- Taking immediate action to rectify problems – which can be quickly and easily fixed on-the-spot, such as clearing blockages and tightening screws, nuts, bolts and fittings
- Checking condition of vehicle instruments – while this is not just a visual inspection it *involves* a visual element in relation to, for example:
  - Confirming travel of pedals and freedom from blockages/interference
  - Activating knobs and switches and *seeing* they function as required.

Vehicle inspection checklist

A vehicle inspection checklist should be used when inspecting a vehicle to record the findings of the inspection.

The following 'Light Motor Vehicle Inspection Checklist' is taken from the 'Light Vehicle Inspection Manual' provided by the Tasmanian Department of Infrastructure, Energy and Resources, Vehicle Operations Branch available at www.transport.tas.gov.au/standards/, (pp. 12 – 13) and provides a sample checklist which may be:

- Used when inspecting a vehicle
- Adapted for use by individual operators to suit their unique requirements and the individual nature of the vehicles they have:

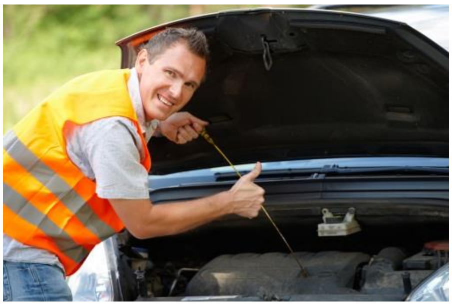
1.	**IDENTITY**	**8.**	**UNDER BONNET CHECK**	**14.**	**EXHAUST**
☐	Compliance plate	☐	Master cylinder	☐	Exhaust leaks
☐	VIN/Chassis No.	☐	Engine oil leaks	☐	Mountings
☐	Engine No.	☐	Engine fuel leaks	☐	Clearance – hoses/wiring
2.	**ELECTRICAL CHECK**	☐	Air cleaner	☐	Catalytic Converter
☐	Headlight/park lights	☐	Engine mountings	**15.**	**REAR SUSPENSION**
☐	Headlight alignment	☐	Exhaust system	☐	Control arms
☐	Tail lights	☐	Battery security	☐	Bushes
☐	Indicators	☐	Wiring security	☐	Ball joints
☐	Brake lights	☐	Steering shaft	☐	Springs/shock absorbers
☐	Reverse lights	☐	Steering coupling	☐	Bump stops
☐	Reflectors	☐	Power steering leaks	☐	Modifications
☐	Lens condition	**9.**	**STEERING**	☐	Suspension height __ mm
☐	Windscreen wipers	☐	Steering box	**16.**	**WHEELS & TYRES REAR**
☐	Windscreen washers	☐	Steering rack	☐	Condition
☐	Horn	☐	Linkages	☐	Tread pattern
3.	**BODY CHECK**	☐	Tie rod ends	☐	Tyre speed rating
☐	Rust	☐	Idler arm	☐	Tyre load rating
☐	Protrusions	**10.**	**FRONT SUSPENSION**	☐	Rims
☐	Bonnet safety catch	☐	Control arms	☐	After market tyres/rims
☐	Windows	☐	Bushes	☐	Wheel track __ mm

☐	Window tint %	☐	Ball joints	**17.**	**FUEL TANK**
4.	**INSIDE CHECKS**	☐	Springs/shock absorbers	☐	Mounting
☐	Heater/demister	☐	Bump stops	☐	Condition
☐	Heater fan	☐	Modifications	☐	Fuel leaks
☐	ABS/SRS lights	☐	Suspension height __ mm	☐	Fuel line
☐	Pedal condition	**11.**	**WHEELS & TYRES FRONT**	**18.**	**LPG GAS**
☐	Brake pedal height	☐	Condition	☐	Gas compliance certificate
☐	Handbrake operation	☐	Tread pattern	**19.**	**BODY & FRAME**
☐	Inhibitor switch (auto)	☐	Tyre speed rating	☐	Rust
☐	Steering wheel	☐	Tyre load rating	☐	Cracks
☐	Steering wheel freeplay	☐	Rims	☐	Bends/kinks
☐	Window operation	☐	After market tyres/rims	☐	Modifications
☐	Door operation	☐	Wheel track __ mm	☐	Mudguards
5.	**MIRRORS**	**12.**	**BRAKES**		
☐	Condition	☐	Pipes and hoses		
☐	Mountings	☐	Discs and drums		
6.	**SEATS**	☐	Calliper/hose security		
☐	Condition	☐	Pads and linings		
☐	Adjustments	☐	Linkages		
☐	Mountings	**13.**	**ENGINE & DRIVELINE**		
7.	**SEAT BELTS**	☐	Mountings		
☐	Condition	☐	Oil leaks		
☐	Compliance	☐	Universal joints		
☐	Mountings	☐	Centre bearing		
☐	Locking/snatch test				

Examples of checks to be made

Regardless of the vehicle type, the following is an indicative and fairly comprehensive list of specific checks which could be carried out when conducting visual inspections of vehicles:

- Checking and topping up fluid levels – as these apply to the individual vehicle in terms of, for example:
  - Engine oil
  - Transmission fluid
  - Water in the radiator and radiator reservoir
  - Windscreen washer water
  - Power steering fluid
  - Brake fluid
  - Clutch fluid
  - Water if the battery



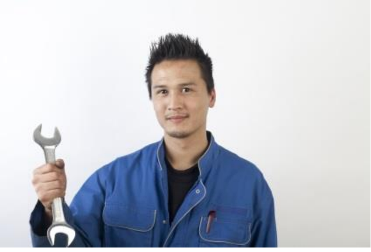
- Checking vehicle tyres – which must address inspecting the on-road tyres and wheels as well as the spare/s, and should involve:
  - Inspecting condition of wheels and rims
  - Checking tyre pressures
  - Verifying all tyres are of the same tread pattern
  - Verifying all tyres are the correct size
  - Verifying all tyres have sufficient tread and sidewalls are not damaged
  - Ensuring on-board tyre repair kits are present and fully stocked
- Checking vehicle lights – to ensure there are no blown globes and all lights are working as intended, inspecting the operation of:
  - Left and right indicators – at the front and rear of the vehicle
  - Brake lights
  - Headlights – on high and low beam
  - Driving lights and fog lights – where fitted
  - Parking/side lights
  - Reversing lights
  - Number plate lights – front and rear
  - Hazard warning lights
  - Interior lights and internal safety lights
  - Dashboard lights and instrument illumination

- Checking all vehicle glass, inside and out – which should look at examining the cleanliness of:
  - Windscreen of the vehicle
  - Rear window
  - Side windows – front and rear
  - Sun roofs – where fitted
  - Mirrors – external/side mirrors as well as internal mirror
  - Check the rubbers on the windscreen wiper blades
- Checking seatbelts fitted to the vehicle – focussing on making sure:
  - Seat belt anchorage points are not cracked, damaged, distorted or rusted
  - No part of the unit is missing
  - The retractor mechanism, locking mechanism, buckle, tongue or adjustment device is working correctly and effectively
  - Seatbelt webbing is not ('Light Vehicle Inspection Manual', Tasmanian Department of Infrastructure, Energy and Resources, Vehicle Operations Branch available at www.transport.tas.gov.au/standards/: (p. 36)
    - Damaged
    - Frayed
    - Stretched
    - Tied in a knot
    - Twisted
    - Split
    - Torn
    - Cut
    - Altered or modified
    - Severely deteriorated
    - Burnt
    - Not correctly and firmly secured to each end fitting
    - Has webbing/stitching which is detached at any point

- Conducting a physical test of the brakes on the vehicle – which should include:
  - Checking the operation of the handbrake
  - Testing the travel of the brake pedal – to ensure it is not low
  - Performing a slow-roll braking test

  See http://smartdriving.co.uk/Driving/Driving_emergencies/Brakes_fail.htm for more and explanation of the above.

- Testing the communication facilities contained in the vehicle – which may include UHF and CB radios I terms of:
  - Sending and receiving
  - Channel selection
  - Control operation/s



- Checking the security features which are included as part of the vehicle and/or have been fitted as 'after-market' additions – such as:
  - Passenger door locks and operation of keys/keyless entry facility
  - Latches and/or locks fitted to luggage compartments
  - Interior locks – such as glove boxes and/or on-board safes
  - Rear and/or side door locks and operation of keys/keyless entry facility
  - Steering locks
  - Vehicle tracking technologies
- Starting the vehicle and running the motor – and:
  - Listening for abnormal engine sounds and/or evidence of exhaust leaks
  - Observing the exhaust for signs of smoke
  - Checking for leaks
  - Making sure the horn works
  - Checking dash board indicators
- Checking maintenance requests/reports from previous trips and vehicle use – to:
  - Identify issues which have been reported by others
  - Determine whether or not necessary repairs/service has been carried out to effectively address the issues
  - Work out if further action needs to be taken to make sure the vehicle is safe/fit for use.

2.2 Complete and retain service reports as required

Introduction

Drivers of Tour Operator service vehicles will be required to complete and retain paperwork that reflects all service work done on those vehicles.

This section identifies points to note when documenting in-house repairs and maintenance performed on service vehicles.

Completing the internal documents

Standard practice as part of the process for providing service and repairs to vehicles is to complete a ‘Service Report’ form/log, ‘Defect Report’ or ‘Request for Service/Maintenance’.

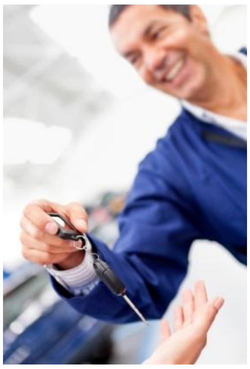
Completing this documentation includes:

- Using the approved template/form – where businesses require this information to be recorded they will provide a document especially for the purpose. The document is commonly hard copy but may be electronic
- Identifying the vehicle which has been serviced/repaired – as required by the business, which requires recording of:
  - Registration number
  - Make and model
  - Odometer reading
- Completing the form immediately after the work has been completed – or as the work progresses, rather than the next day or at the end of the week
- Describing service and repairs provided including date – in sufficient detail to allow others to interpret and understand the work that was undertaken.

  Keys are:
  - Write clearly
  - Be honest
  - Cover all work done
  - Use recognised abbreviations and acronyms
  - Cross-reference to parts used
- Noting matters that will/may require attention at the *next* service – so that:
  - Vehicle users/operators are aware of this – and can factor it into their use of the vehicle
  - Management/administration are made aware of significant/costly issues which are imminent
  - Necessary resources, parts and other requirements are purchased/obtained in advance of the next scheduled service session

- Recording the action taken for cost and stock control purposes – which may require:
  - Listing items used – parts and fluids
  - Costing each unit/item
  - Including hours spent on the job
  - Calculating totals – for parts and labour involved in service and/or repair provision
- Updating *next* scheduled service for the vehicle – in terms of (as appropriate):
  - Date when service is required
  - Odometer reading when service is required

  This update might be required:
  - in a central Service Index – kept to monitor service dates for all vehicles
  - In the manual for the vehicle – kept in the vehicle or office
  - On a sticker placed on the windscreen of the vehicle



- Attaching the inspection checklist (or copy) which initiated/triggered the work to the completed 'Service Report' – if required
- Returning the vehicle to its normal location – to return it to service
- Notifying personnel who may be waiting on the vehicle – that it is now ready
- Processing paperwork – forwarding it as required to administration/management.

Websites

See:

- download.microsoft.com/download/9/d/c/.../Vehicle_Repair_Log.xlt – Vehicle repair log
- http://www.vertex42.com/ExcelTemplates/vehicle-maintenance-log.html - Vehicle maintenance log.

2.3 Adhere to scheduled maintenance and service requirements as set out by the vehicle manufacturer

Introduction

Maintenance to service vehicles must be provided in accordance with requirements provided for each individual vehicle.

This section looks at sample scheduled maintenance provisions detailing actions which may be required.

Generic requirements

The exact requirements for each vehicle will vary between makers and models of vehicles – the following is indicative of advice/directions which may be covered in Vehicle Manuals/Operating Instructions:

- Vehicles must be serviced at nominated times – as appropriate, designated by either:
  - Distance travelled, or
  - Operating hours
- Scheduled maintenance and service may need to be undertaken by specified businesses in order to preserve vehicle warranties – these may be:
  - The organisation from whom the vehicle was purchased
  - An authorised service dealer

- Internal protocols and/or host country legislation may require safety critical maintenance and service (such as work on brakes, steering and suspension) on vehicles to be carried out by suitably qualified mechanics – and not performed internally by non-certified employees
- Stated checks must be performed as and when prescribed in the vehicle manual – these can include:
  - Visual inspections of listed components
  - Operating checks of listed components
  - Nominated testing of systems, items and on-board technologies.

Sample requirements

Note: the following is an indicative list of the work that may need to be carried out and it is not a comprehensive/exhaustive list.

When providing maintenance and service to vehicles there can be a need to:

- Check operation of all dashboard gauges
- Run checks on all OBD devices

- Perform physical check of all pedals, knobs, switches and actuating devices – wipers, lights, indicators, radios, horn
- Grease all lubrication points
- Top-up or drain and replace engine oil
- Top-up or drain and replace gearbox oil
- Top-up or drain and replace power steering fluid
- Top-up or drain and replace hydraulic clutch reservoir
- Bleed and replace brake fluid
- Check battery condition
- Top-up or drain and replace transmission fluid
- Top-up or drain and replace differential oil
- Check brake wear and replace parts where necessary
- Clean and/or replace nominated filters – air, oil and/or petrol
- Add coolant to radiator
- Check/replace radiator hoses
- Check condition of wheels
- Check tyres:
  - Pressure
  - Tread depths
  - Rotate tyres
  - Check spare/s
- Check condition of seats belts
- Check chassis for cracks
- Inspect suspension components
- Check/replace all belts
- Check/replace plugs
- Tune engine.

2.4 Conduct special checks and service after designated trips

Introduction

Tour Operators commonly require vehicles to be checked, and where necessary, serviced after certain designated trips.

This section provides a context for this requirement and details activities involved in the inspection process.

Context

Checks made of vehicles after trips are referred to as ‘post-trip checks’.

These checks:

- Are as important as the pre-departure checks
- Need to be undertaken to provide the basis for maintaining the vehicle in a serviceable condition – and extend the life of the vehicle and optimise its re-sale/trade-in value
- Need to be recorded on a designated form – the same form used for pre-departure checks is often used for this activity too
- May only be undertaken after certain trips/uses – such as:
  - After vehicle has been used in severe conditions, including extended periods in dust, dirt, mud and sand, water crossings
  - After long periods in low gears or long periods where four-wheel drive was engaged
  - After significant times in mountainous terrain
- Always include an internal sweep of the vehicle to check for property left behind by drivers/passengers
- Regular periodic checks on tool kit and on-board spare parts carried by each vehicle to ensure all items are present as required.

Post-trip checks

Post-trips checks should embrace:

- Internal checks
- External checks.

Internal checks

Inside the vehicle attention should be paid (as/if necessary) to:

- Checking to for luggage/property which has been left inside the vehicle, door pockets, glove boxes, the boot or any other vehicle spaces
- Removing loose debris from floor, seats, bins and similar – see ‘Maintain tourism vehicles in a safe and clean operational condition’

- Sweeping and cleaning vehicle trays – see 'Maintain tourism vehicles in a safe and clean operational condition'
- Vacuum cleaning and spot cleaning carpeted floors and other surfaces – see 'Maintain tourism vehicles in a safe and clean operational condition'. Always directions/recommendations in the Vehicle Manual when cleaning vehicle interior, especially when wet cleaning
- Cleaning upholstery – see 'Maintain tourism vehicles in a safe and clean operational condition'
- Cleaning door jambs and steps – see 'Maintain tourism vehicles in a safe and clean operational condition'
- Cleaning all inside glass and mirrors – see 'Maintain tourism vehicles in a safe and clean operational condition'
- Cleaning the steering wheel, dashboard, vehicle controls and consoles – see 'Maintain tourism vehicles in a safe and clean operational condition'
- Completing necessary paperwork – which may include:
  - 'End of Trip Vehicle Inspection Checklist'
  - 'Lost and Found Report'.

See more at:

- https://www.youtube.com/watch?v=t--JMt9cECU – Interior auto detailing
- https://www.youtube.com/watch?v=rT86aCFUDUc – How to clean upholstery: hot water extraction
- https://www.youtube.com/watch?v=dag7N8me1tE – Door jamb detailing and cleaning: how to clean door jambs to perfection!
- https://www.youtube.com/watch?v=QY6_3o6rpLc – Detailing black door jambs: Darren's professional tips!
- https://www.youtube.com/watch?v=aVUtdlzrktI – Car glass: Cleaning and polishing
- https://www.youtube.com/watch?v=hq671_hfhFE – How to streak free clean windows: Auto detailer's tip
- https://www.youtube.com/watch?v=LvrkassME74 Cleaning and detailing car dash
- https://www.youtube.com/watch?v=S1pe5JXqN1o – How to clean & protect vinyl, rubber and plastic
- https://www.youtube.com/watch?v=nsBVnuOtr6U – Cleaning vehicle interiors
- https://www.youtube.com/watch?v=qpryIxF0hiY – Detailing a car auto interior cracks and crevices.

External checks

Outside the vehicle attention should be paid (as/if necessary) to:

- Using a pressure washer to clean external surfaces and vehicle wheels – see 'Maintain tourism vehicles in a safe and clean operational condition'.

  Have a look at:

  - https://www.youtube.com/watch?v=kRQ3QilQtJA – Pressure washer basics
  - https://www.youtube.com/watch?v=ONm0DL5koRg – How to wash a car engine.

- Cleaning the outside windows and mirrors – see 'Maintain tourism vehicles in a safe and clean operational condition'.

  See more at:

  - https://www.youtube.com/watch?v=Ob-hN0LZbbQ – Cleaning auto glass
  - http://www.supercheapauto.com.au/how-to/Car-Cleaning/Cleaning-Windscreen-and-Windows.aspx?id=100 – Cleaning windscreen and windows
  - https://www.youtube.com/watch?v=D05vX-G9iV4 – How to super clean your windshield
  - https://www.youtube.com/watch?v=m2B57fxQSP4 – RainX windshield rain repellent test and review

- Polishing chrome and fittings – see 'Maintain tourism vehicles in a safe and clean operational condition'
- Applying tyre black – see 'Maintain tourism vehicles in a safe and clean operational condition'.

2.5 Check performance and efficiency of vehicle during operation

Introduction

A standard requirement for vehicle drivers is the need for them to monitor the performance and efficiency of the vehicles they drive.

This section describes the practices that need to be commonly implemented by drivers in this regard.

Ways to monitor vehicle performance and efficiency

Generally the following activities need to be applied:

- Monitoring on-the-road vehicle operation – this means while driving vehicles the driver needs to be on the lookout for:
  - Unusual noises or sounds that are not normal or indicate a problem
  - Unusual or excessive vibrations
  - Unusual smells or odours that may indicate components are getting too hot
  - The presence of smoke
  - Equipment that is malfunctioning
  - Intermittent faults with equipment, items or technology/systems
- Checking fuel consumption of the vehicle (see below) and comparing it against distance travelled – taking into account:
  - Type of terrain covered – hilly/mountainous, on-road or off-road conditions
  - Load/weight of the vehicle with reference to passengers and cargo carried
  - Gear which was primarily being used
  - Type of traffic – 'stop-start' or highway travelling
- Recording issues that require attention – which will usually require comments/notes to be physically written down:
  - In individual vehicle log books and/or trip record books
  - On an electronic tablet/device
- Reporting issues which have been identified to the appropriate person – which will typically require:
  - Verbal reporting of an issue to a supervisor or the maintenance department – as part of standard reporting protocols
  - Submission of the competed paper-based report to a nominated internal person – such as an administration person, vehicle manager or similar

- Obtaining scheduled maintenance for the vehicle while on the trip, as/if required – the majority of trips will *not* necessitate service/maintenance as the vehicle will usually be suitably prepared prior to the trip.

  This said, on long trips there can be a need to obtain scheduled service when on very long trips or to obtain service to effect repairs when damage has been incurred.

  Obtaining such service may involve:

  - Pre-arranging *scheduled* service with providers prior to departing on the trip – by establishing (for example) a line of credit with the service provider and making a date/time for the service
  - Contacting head office when damage needs to be repaired so they can make enquiries on behalf of the driver and the Tour Operator to:
    - Identify a suitable repairer
    - Make arrangements for the work to be done
    - Arrange payment details.

Calculating fuel consumption

Calculating fuel consumption for a vehicle is a relatively simple exercise.

The same approach/equation is used regardless of whether the calculation seeks to work out fuel consumption in terms of:

- 'Kilometres per litres' or 'miles per gallon'
- Petrol or diesel.

To calculate fuel consumption (in terms of kilometres per litre) simply divide distance travelled by fuel used.

For example:

- Distance travelled = 875 kilometres
- Fuel used = 294 litres
- 875 ÷ 294 = 2.98 km/L.

Websites

For online calculators which automatically calculate fuel consumption when the appropriate figures are entered, see:

- https://motormouth.com.au/myvehicle/consumptioncalculator.aspx
- http://calculator-converter.com/l_100km_calculate_l_100_km_fuel_efficiency_consumption_economy_calculator.php.

2.6 Undertake minor routine repairs

Introduction

It can be necessary for drivers to undertake a range of minor repairs when they are on-the-road to maintain the effective operation of the vehicle.

This section describes indicative work needed to complete several of the common requirements in this regard.

Important notes

In relation to this section it is vital those undertaking repairs note:

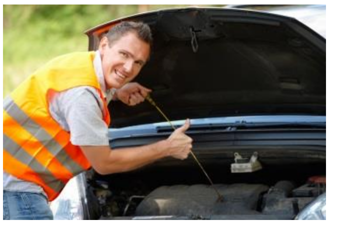
- Performing repairs to vehicles which are still under warranty can void the warranty – so there is a need to check this before proceeding any further
- The actual work required may differ between makes and models of vehicles – the procedures listed in the notes are **indicative only**
- Always read Owner's manual or Workshop manual to determine exact specifications for individual vehicles
- Obtain the right parts before starting the job
- Obtain and use the right tool/s for the job
- Work steadily – never rush a job
- Check the surrounding areas/sections for debris and/or damage at the same time – and respond as indicated
- Never be afraid of asking for help, advice or direction from a more qualified or experienced person or head office.

Replacing blown globes

Examples of globes which may need to be changed

There can be a need to replace blown globes for:

- Turning lights
- Brake lights
- Low and high beam headlights
- Driving lights
- Interior lights
- Internal safety lights
- Reversing lights.



Indicative instructions for changing a blown globe

To replace a light globe:

- Identify the light to be replaced
- Obtain correct globe for the job – sealed units will require the entire headlamp to be replaced; semi-sealed units only require the light globe
- Remove the light cover/dust covert
- Unplug electrical connector:
  - In some cases there is a need to remove the entire light assembly to replace a light globe
- Disconnect whatever mechanism is used to keep the old globe in place
- Remove the old globe noting how it sits – bayonet globes are removed by rotating them anti-clockwise

- Put the new globe in position taking care not to touch the glass part of the globe with the fingers
- Re-attach the mechanism, electrical connection and cover
- Test the light to make sure it is working.

Websites

See the following for more help:

- http://www.supercheapauto.com.au/how-to/Outside-the-Car/Checking-and-Changing-an-Automotive-Light-Globe.aspx?id=79 – Checking and changing an automotive light globe
- https://www.youtube.com/watch?v=6hURsa46t4M – How to change a headlight bulb
- https://www.youtube.com/watch?v=jk1kFc56dwk – How to replace headlight bulbs
- https://www.youtube.com/watch?v=cB99QN1AYBU – How to install replace front lower side reflector
- https://www.youtube.com/watch?v=mgGWFqPe0Ms – Headlight bulb replacement service: 2005 Honda CRV
- https://www.youtube.com/watch?v=5F31wzh60zo – How to replace taillight
- https://www.youtube.com/watch?v=9POigqpLxQc – How to replace fog light and bulb.

Replacing a fan belt

Indicative actions are:

- Release tension on old belt at bolt on tensioner rail
- Move alternator to allow free play in old belt
- Remove old belt
- Fit new belt by looping it over alternator and water pump pulleys
- Re-locate the alternator to bring tension on the new belt
- Tighten the bolt on the tensioner rail

- Run the engine to test for:
  - Squealing noise – indicating belt is slipping and needs to be re-tensioned
  - Dashboard warning lights – which may indicate a problem.

Websites

See also:

- https://www.youtube.com/watch?v=4hnOUs9eiN4 – Change a fan belt
- http://www.supercheapauto.com.au/how-to/Servicing-Mechanical/Replacing-a-Drive-Belt.aspx?id=76 – Replacing a drive belt
- https://www.youtube.com/watch?v=Wg1AX77xEBQ – How to replace a serpentine belt
- https://www.youtube.com/watch?v=01J3Fp9_bMA – How to replace a serpentine belt: Toyota Corolla
- https://www.youtube.com/watch?v=XD0LNkFYPGQ – Toyota timing belt
- http://www.wikihow.com/Change-an-Automotive-Belt.

Replacing radiator hoses

Steps are:

- Drain radiator – if necessary, using drain plug if fitted: there may be a requirement to retain the fluid to put back in the radiator
- Remove bottom radiator hose from radiator if there is no drain plug
- Ensure engine has cooled before removing radiator cap
- Use pliers to release clamp holding hose in place – at front and back of top and bottom hose (if changing both hoses). It may be a good time to replace the thermostat while the top hose is off the vehicle
- Use flat screwdriver to help loosen connection and twist hose to break seal and remove the hose
- Clean the parts to which the hose ends were secured – to remove visible debris
- Determine if new hose clamps need to be used to fit the new hose/s – use new clamps where indicated
- Position clamps (using pliers to open them up) near end of new hoses and replace hoses
- Re-attach clamps
- Fill radiator
- Run and check for leaks.

Take the time to view:

- https://www.youtube.com/watch?v=AGmQ50naiGg – How to replace a radiator hose (upper and lower)
- https://www.youtube.com/watch?v=C64z0L9816A – Replacing radiator hose
- https://www.youtube.com/watch?v=xH0D22ITjQU – How to replace the upper radiator hose.

Replacing spark plugs

Steps are:

- Disconnect the HT lead to the spark plug
- Use a spark plug socket attached to a wrench in an anti-clockwise direction to loosen the used spark plug – many plugs can be hard to shift so a long handle will provide the necessary extra pressure
- Use the fingers to unscrew the old spark plug and remove it once it has been freed
- Take care not to allow any debris to fall into the cylinder
- Clean the plug with a wire brush and check the condition of the plug to determine:
  - If a new plug is required
  - If there is evidence of another problem which should be investigated
- Use a feeler gauge to set the correct gap for the old or new spark plug to be fitted
- Use fingers to re-seat the plug into the block
- Tighten gently until finger tight and ensure thread is not crossed
- Finish tightening with spark plug socket attached to a torque wrench to required torque specification
- Re-attach HT lead
- Start and run motor to test.

See also:

- http://www.supercheapauto.com.au/how-to/Servicing-Mechanical/Replacing-Spark-Plugs.aspx?id=68 – Replacing spark plugs
- https://www.youtube.com/watch?v=jj06A6UdHvI – Changing spark plugs and replacing plug wires.

Replacing glow plugs

Indicative actions are:

- Remove any visible debris around the plugs which may fall into the cylinders – use of compressed air is recommended
- Disconnect the (negative) battery terminal
- Use a spanner to remove the nuts holding the bridging strap to the plugs
- Remove the bridging strap
- Use a spanner to untighten the glow plug and remove with fingers
- Test the new glow plug to be fitted
- Ensure plug and thread is clean
- Screw new plug in by hand until seated in cylinder head
- Tighten with torque wrench to torque specification according to the individual plug

- Replace bridging strap securing with nuts
- Re-connect battery
- Test.

For more information see:

- https://www.youtube.com/watch?v=RCcmN-r5Z9M – Replacing glow plugs in a Mitsubishi Delica L400
- https://www.youtube.com/watch?v=4Po_1pNHE1Y – Glow plug change
- https://www.youtube.com/watch?v=4Po_1pNHE1Y – Easy and complete glow plug test.
- http://autorepair.about.com/od/enginerelateddiyjobs/l/aa081404a.htm - How to replace your diesel glow plugs.

Replacing a fuse

Indicative steps are:

- Locate fuse box/block
- Remove cover
- Read key on cover (or in manual) to determine the fuse to replace
- Remove fuse to be replaced
- Insert new fuse – press in position with finger
- Verify spare fuses are available for emergency use – where space provided to store them
- Test vehicle/accessory/light which was operating on this fuse. If fuse blows immediately then this usually indicates another problem (a short) and the vehicle may need to be taken to an auto-electrician.
- If light/accessory functions as required, replace fuse box/block cover.

See more at:

- https://www.youtube.com/watch?v=kOXeo-kqg68 – How to replace a blown fuse in a car
- https://www.youtube.com/watch?v=faWZmG0fXy8 – How to find and replace a blown fuse in your car or truck.

Replacing mirrors

Indicative actions for replacing side mirrors are:

- If mirrors are simply attached by screws or bolts – remove the screws or bolts, remove the old mirror, locate the new part and re-attach the screws or bolts
- Where mirrors are fitted to the front frame (left or right-hand side) corner of the frame:
  - Remove internal trim by gently levering cover away
  - Where mirrors are powered there is a need to disconnect the battery and can be a need to remove inside door handles, trim panel and plastic cover/sheet – disconnect the wiring connector for the mirror

- Remove grummets and undo the bolts connecting the unit/mirror to the frame, supporting the mirror as this is done
- Remove the old unit
- Connect new unit fastening bolts and attaching wiring connector
- Replace trims, panels and items removed during disconnection
- Re-connect battery
- Test.

Websites

See:

- https://www.youtube.com/watch?v=U-1y8NIEhKM – Auto side mirror replacement
- https://www.youtube.com/watch?v=77lBQ_CRqdo – Replace your passenger side mirror
- https://www.youtube.com/watch?v=d8ASeN4rgek – Plastic backed mirror glass installation
- https://www.youtube.com/watch?v=D6j8fwGCrHs – Side mirror glass replacement.

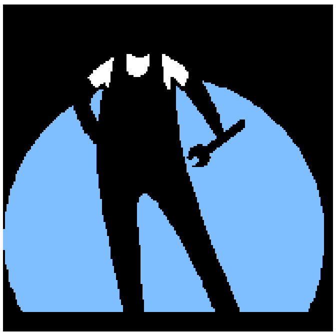
Alignment of external mirrors

It is necessary for driver's to ensure their external mirrors remain correctly aligned so there are no 'blind spots'.

As the alignment of mirrors will depend on the driving position of the driver, alignment of mirrors should be undertaken every time there is a change if driver or the driving seat position is altered.

Checks should ensure:

- Mirrors can be moved by hand – where applicable
- Remote controls function correctly to move mirrors as required.



Vehicles will generally feature:

- One internal, rear-view mirror
- Two external mirrors – one on the left-hand side of the vehicle and one on the right-hand side.

See:

- https://www.youtube.com/watch?v=QpRovzCeM4Q – Are your car mirrors adjusted correctly?
- http://www.wikihow.com/Set-Rear%E2%80%90View-Mirrors-to-Eliminate-Blind-Spots.

Changing tyres

See the videos at:

- https://www.youtube.com/watch?v=fGp1meWKCD8 – How to change a tire
- https://www.youtube.com/watch?v=joBmbh0AGSQ – How to change a tire: change a flat car tire step-by-step.

Repairing punctures

Most operators have an SOP whereby all punctures are taken to a tyre dealership/garage to have them repaired.

Check the videos at:

- https://www.youtube.com/watch?v=ulQmDaqv4F4 – Best tire repair video
- https://www.youtube.com/watch?v=LRQJgTxRhOA – How to repair a puncture in a tyre with a repair plug/string
- https://www.youtube.com/watch?v=tmW9YJpVwO4 – MSCTC Tire changing training video [using a commercial machine]
- https://www.youtube.com/watch?v=2QR1vag1N9Q – Professional tire repair
- https://www.youtube.com/watch?v=wi5uBUaMsrA – How to repair a car or light truck tire.

Tyre inflation

When punctures have been repaired the tyre may need to be inflated.

This may be achieved using:

- An external pump powered by the electrics of the vehicle
- Tyre inflator at a garage/service station.

Work Projects

It is a requirement of this Unit you complete Work Projects as advised by your Trainer. You must submit documentation, suitable evidence or other relevant proof of completion of the project to your Trainer by the agreed date.

2.1 To fulfil the requirements of this Work Project you are required to nominate a vehicle (to be approved by your Assessor) and monitor that vehicle providing evidence you have:

- Performed **a range of** regular preventative maintenance and operational checks of the vehicle as agreed with the Assessor
- Completed and retained service reports
- Complied with required scheduled servicing requirements
- Conducted special checks after designated trips
- Checked performance and efficiency of the vehicle during operation
- Undertaken **a range of** minor routine repairs.

Summary

Monitor service vehicles

When monitoring service vehicles:

- Undertake preventative maintenance as scheduled for each vehicle as designated
- Arrange for external service where required
- Check vehicles as required prior to use
- Realise the importance of cleaning in the monitoring and maintenance processes
- Use a checklist to guide and record vehicle inspections
- Perform visual and operational checks as part of standard inspections
- Complete required service reports for all work undertaken on vehicles
- Retain and/or otherwise process (according to organisational requirements) all documentation relating to vehicle service
- Carry out special checks on vehicles after nominated trips where conditions were harsh and/or distances were long
- Maintain a check on the performance of each vehicle to determine its fuel efficiency
- Undertake minor routine on-the-road repairs as necessary to maintain operational status of the vehicle/s.

Element 3: Monitor road conditions

3.1 Assess road conditions

Introduction

All drivers need to assess road and traffic conditions on an ongoing basis.

This section identifies reasons to assess *expected* road and traffic conditions, and provides advice on what needs to be done in this regard prior to setting out on a trip.

The following section (section 3.2) covers activities required to assess actual road and traffic conditions when on-the-road.

Need to monitor and assess

Road and traffic conditions which need to be travelled should be monitored and assessed so:

- An idea of the conditions to be expected can be obtained – so the driver can start to get a mental picture of the up-coming trip
- The optimum route can be selected where the driver has the option of picking roads – rather than being bound by an itinerary or pre-prepared route. Sometimes, the optimum route may be:
  - The shortest route in terms of distance
  - A lengthier route but one which can be travelled in a shorter time
  - A route that includes certain sights, destinations or attractions
  - A path that avoids known obstacles, delays or problem situations

- An alternate route can be chosen if necessary – where unacceptable roads or conditions are identified
- Driving times can be calculated – based on:
  - Distances to be travelled
  - Types of terrain and roads which are expected
  - Local laws regarding driving times and rest breaks for drivers
  - Plans for the trip, itineraries or trip-related schedules
- The driver can decide which vehicle type needs to be taken – based on expected terrain and conditions.

Activities required

Activities required to assess road and traffic conditions when on-the-road include:

- Checking the route before commencing journey – this should include looking at a map and/or talking to other drivers and considering identification of:
  - Potential hazards – with reference to:
    - Bad/sub-standard roads
    - Road work
    - Difficult terrain – such as steep climbs and descents, unsealed roads, rough off-road conditions
    - Long distances where there is no fuel available
    - Changing road or travel conditions

  - Weather to be expected along the way – which may include:
    - Visiting relevant metrological websites
    - Contacting destinations
    - Speaking with other drivers
  - Expectations regarding traffic to be encountered – in terms of:
    - Determining expected levels/volumes of traffic which might be encountered
    - Considering expected type of traffic likely to be using the road at the time
    - Considering the direction of moving traffic – for example, will the journey be 'with' or 'against' the main flow of traffic
    - Local events which are occurring along the route and which are likely to generate higher levels of traffic on the road
- Obtaining details regarding road conditions – which may necessitate:
  - Telephoning road authorities
  - Visiting websites of motoring bodies
  - Contacting emergency authorities
  - Talking to other drivers and/or destination or local attraction contacts
- Factoring in potential support available along the proposed/required route – which should address consideration of:
  - Location and availability of fuel
  - Location and availability of suitable providers for repairs and maintenance
  - Emergency services that could be called on for assistance in the event of an accident.

3.2 Assess traffic conditions

Introduction

All drivers must constantly monitor and assess the road and traffic conditions they are encountering when driving.

This section looks at the need to monitor and assess on-the-road conditions, and provides advice on evaluating traffic conditions when on-the-road.

Need to monitor and assess

Road and traffic conditions need to be continually monitored and assessed when on-the-road so:

- Driving can be adjusted to reflect the conditions
- Passengers receive a safer and smoother ride
- The optimum path of travel can be selected on the road
- A suitable CAS can be maintained around the vehicle
- An alternate route can be chosen if necessary
- Information about the observed conditions can be communicated to Head Office and/or shared with other drivers.

Evaluating traffic conditions

When driving there is a need to:

- Be aware of the volume of traffic on the road – both travelling in the same direction and on-coming
- Judge need to slow average travel speed – to align with changed/changing conditions
- Note the proximity of other vehicles – in terms of:
  - Vehicles in front and behind
  - Vehicles in adjacent lanes
  - Oncoming vehicle position and number
  - Parked and stopped vehicles
- Assess the speed of other road users
- Be familiar with the type of other vehicles using the road
- Use the extra height of the vehicle (where applicable) to look above cars/other traffic to 'see ahead' – in order to:
  - Determine traffic flow patterns
  - Identify blockages on the road and the upcoming need to slow and/or stop the vehicle.

3.3 Feedback information to head office about road and traffic conditions

Introduction

Standard practice in the industry is for drivers to provide information back to head office and other drivers regarding actual road and traffic conditions they are experiencing on-the-road.

This section explains why this is done highlighting the benefits this action can bring to the company, other drivers and ultimately the Tour Operator.

Ways to provide feedback

Feedback about road and traffic conditions is usually provided:

- Using the two-way radio on board the vehicle
- Using driver cell phone – in compliance with local laws regarding use of these when driving a vehicle.

Need to feedback information

Drivers need to feedback information back to head office and other drivers regarding actual road and traffic conditions they are experiencing on-the-road in order to:

- Keep head office up-to-date regarding their progress – which may be important for the purposes of them:
  - Notifying waiting passengers
  - Scheduling other services
  - Tracking progress
  - Contacting venues/destinations to advise of delays

- Provide up-to-date advice for other drivers – so:
  - They can benefit from the most recent experience of an area/route or road
  - They can avoid trouble-spots and other situations likely to cause delays – see next section for examples
  - They gain an understanding of the amount of traffic
  - They can plan an alternate route
  - They can adjust their expected driving time/s
- Give road and weather reports – about:
  - Conditions of roads as a result of past weather, especially rains, storms and floods
  - Current weather conditions
  - Forecasted weather conditions based on personal observation.

3.4 Identify factors that may cause delays or route deviations

Introduction

There is always the potential for vehicles to be delayed for some reason or to have to take an alternate route.

This section identifies common causes which may cause a delay or necessitate deviating from a planned route in order to avoid or accommodate a problem that has been encountered.

Factors with the potential to cause delays

Despite the best planning and pre-departure checking there will always be situations where a planned route needs to be varied.

Any of the following are likely to cause a planned trip to be delayed or give rise to the need to generate a deviation to the one which was originally planned:

- Vehicle-related problems – including:
  - Traffic accidents which have blocked the road or closed one or more lanes of traffic
  - Vehicle breakdowns
- Floods – which:
  - Have closed roads
  - Are making roads impassable – even though they have not been officially closed
  - Appear imminent based on local rainfall/conditions
- Restrictions on road usage placed on vehicles over stated dimensions – as this applies, for example, to long vehicles (say, utilities towing trailers) and/or wide vehicles:
  - On roads where there are sharp/tight bends
  - On narrow roads
- Weather related damage to roads – such as:
  - Severe potholes
  - Wash-aways – roads and bridges
  - Landslides
  - General damage to bridges and/or tunnels
- Road works – including:
  - Repairs and re-surfacing
  - Line marking
  - Digging up of roads caused by repairs to utilities such as gas/water

- Building construction – which may give rise to:
  - Parking restrictions in local areas
  - Closure of one lane of traffic for cranes and other construction vehicles
  - Interruption to traffic flow at the site caused by vehicles entering and leaving the construction site
- Emergency situations impacting on road use – such as:
  - Police operations
  - Official road blocks
  - Bush/wildfires
  - Building fires
  - Chemical spills
  - Fuel leaks
- Road closures due to local, special events – such as:
  - Festivals
  - Marches
  - Sporting events
  - Parades
  - Religious ceremonies
- Unusual levels of traffic – which may be the result of:
  - Holiday traffic
  - Weather
  - Other routes being closed/slow.

3.5 Factor conditions into choice of route to be taken

Introduction

As the two previous sections have indicated, if conditions road and/or traffic conditions change there can be a need for drivers to revise their planned route and select an alternate one.

This section addresses actions and considerations drivers may need to take into account when a decision has to be made to alter a pre-determined route.

Actions and considerations

When a need arises to choose another way to get to a destination or attraction, the following need to be addressed:

- Safety is always the primary consideration – this means a driver must not place the safety of any passengers in the vehicle at risk if a hazard has been identified.

  Examples of what may be seen as unacceptable risk can include:
  - Trying to drive a vehicle across a flooded road, river or causeway – where the all too common consequence is the car/vehicle (even 4WD vehicles) is washed away by the flow/force of the water
  - Driving at night in some situations – see below
  - The driver spending 'too long' behind the wheel without adequate rest breaks
- Taking a shorter way than the road identified on the itinerary or the plan for the trip – where:
  - There is a risk a non-negotiable deadline has to be met – such as:
    - Providing catering/meal service to a tour group
    - Arriving at a venue to meet their check-in times for accommodation
    - Transferring passengers to another carrier who has a scheduled departure time
  - Late arrival at a destination will not allow passengers sufficient time to enjoy their experience at that location
  - The vehicle needs to return to base by a given time – in order for the next scheduled use of the vehicle
- Safety of the vehicle – the Tour Operator will always expect the driver to take good care of the vehicle which can mean:
  - **Driving at night** in an area known to be teeming with wildlife thus posing a very high risk of hitting an animal is unacceptable – so the driver may need to:
    - Stop and find accommodation
    - Realise this potential situation sufficiently far in advance that a shorter route can be driven to get to the required destination in daylight
  - Choosing a better quality road where the road condition/surface is likely to cause damage to the vehicle

- Keeping people in the vehicle informed about what is happening – and:
  - Explaining the changes that are going to take place
  - Giving reasons for those alterations
  - Apologising, where necessary/appropriate

- Revising scheduled/advertised inclusions to ensure next or final destination is reached on-time – which may involve:
  - Reducing time spent at certain/each destination or attraction
  - Removing one or more destinations/venues from the trip/tour altogether
  - Giving people who are on-board shorter rest breaks/stops to make up time.

3.6 Read a map

Introduction

Despite the presence of GPS and various in-vehicle navigation systems it is useful if drivers know how to read and use a map.

This section identifies the basic steps involved in map reading which should be practiced so that competency can be gained.

Basic requirements

- Use a dedicated **road** map – which will be best for showing locations, distances, road classifications and relevant other travel-related information
- Obtain the latest version of the map for the area – as this will be the one which best reflects new roads, place names and changes to the layout of areas
- Never read the map when driving – if there is a need to read a map when on a trip:
  - Stop the vehicle to read the map
  - Read the map during scheduled stops/while at a destination

- Check out the map and learn its features – in terms of:
  - The scale used on the map to show distances
  - How different classes of roads are indicated
  - How distances between points are shown
  - Whether distances are in kilometres or miles
  - Symbols used to indicate landmarks/points of interest.

Using the map to determine a route

To work out the shortest way between two points, standard action includes:

- Identifying point of departure/present location of the vehicle
- Locating the required destination point
- Drawing an imaginary (or actual) straight line between the two points – this is the shortest line between the two points
- Looking at the map to determine suitable roads which approximate the imaginary (or actual) straight line between the two points – subject to:
  - Ensuring roads are of a suitable standard – as opposed to being tracks, or roads which are (say) unsealed
  - Making sure the proposed route does not involve unacceptable conditions – such as river crossings, climbs of descents which are 'too steep' or travel through towns which may impede travel time
- Writing down a list of destinations and distances – to use as a checklist when driving to ensure the selected route is followed.

Work Projects

It is a requirement of this Unit you complete Work Projects as advised by your Trainer. You must submit documentation, suitable evidence or other relevant proof of completion of the project to your Trainer by the agreed date.

3.1 To fulfil the requirements of this Work Project you are required to nominate a geographic area and time period (to be approved by your Assessor) and monitor road conditions within those parameters providing evidence you have:

- Assessed road conditions
- Assessed traffic conditions
- Communicated information about road and traffic conditions to others
- Identified factors that may cause delays or route deviations
- Factored conditions into choice of route to be taken for a nominated trip.

3.2. To fulfil the requirements of this Work Project you are required to read a map and explain (verbally or in writing) how to get from one location to another.

Summary

Monitor road conditions

When monitoring road conditions:

- Take time to research the route to be driven in order to assess possible road and weather conditions which may be encountered
- Assess traffic conditions on an ongoing basis while on-the-road
- Adapt driving to match/reflect the traffic and road conditions actually encountered
- Seek advice and input from other road users
- Provide advice and information about road and traffic conditions to Head Office and others
- Be alert to identifying factors which may necessitate a variation in proposed route
- Be prepared to determine a better route when facing an obstacle or delay
- Use a map (or GPS technology) to help determine alternate routes and/or roads to travel.

Element 4: Drive vehicles

4.1 Demonstrate basic operational driving practices

Introduction

There is a need for all drivers to become competent in basic operational driving practices as a pre-requisite for taking the practical element of their driving test.

This section provides advice to support on-the-job training as well as training provided by licensed driver training schools/centres.

Standard driving

It is important to understand basic operational driving practices must include:

- Compliance with host country driving rules and regulations
- Demonstrated competency under normal driving conditions and touring conditions
- Realisation a licence to drive a car does not entitle the holder to drive a bus/coach.

In order to achieve the above, the following is indicative of what is required:

- Learning and obeying the road rules, road markings, road signs and traffic signals/controls
- Getting lots of supervised practice
- Controlling the vehicle correctly – in terms of:
  - Conducting pre-drive checks of controls/mirrors
  - Starting the vehicle and moving off – including:
    - Undertaking 'head checks' to gain situational awareness
    - Visually scanning the road, traffic, pedestrians and general conditions which impact travel
    - Maintaining a safe space (CAS) around the vehicle
  - Accelerating and braking – which directly relates to controlling the pedals of the vehicle
  - Making safe driving decisions and responding appropriately to identified risks and hazards
  - Turning left and right, maintaining lane and changing lanes – which directly relates to controlling the steering wheel of the vehicle
  - Reversing
  - Parking
  - Using the knobs and other control features of the vehicle

- Anticipating potential hazards and managing risky situations – with specific reference to understanding the dangers/risks associated with:
  - Driving when affected by alcohol and/or drugs
  - Speeding
  - Using cell phones and other mobile devices
  - Driving with passengers/other people in the vehicle who may pose a distraction
  - Driving when tired
  - Not wearing seatbelts
  - Failing to ensure the vehicle has been properly maintained in a safe and roadworthy condition

- Driving the vehicle on flat ground, up hills and down hills – including being able to do hand-brake starts on hills
- Being aware of 'blind spots'
- Obtaining significant on-road practical experience under a variety of road, traffic, time (day and night) and weather conditions
- Gaining general confidence in driving and how to control the vehicle in a variety of conditions
- Concentrating on the job of driving safely and adapting driving style/speed to reflect and adjust to emerging conditions.

Websites

Take time to view:

- https://www.youtube.com/watch?v=K6fT1D2-YYI – Driving safely: defensive driving in a dangerous world
- https://www.youtube.com/watch?v=tBlm4AV4ovw – Car driving safety tips: Auto Care Series
- http://www.qld.gov.au/transport/licensing/getting/education/videos/ - Safe driving videos, including:
  - Pre-drive vehicle safety check
  - Getting comfortable in the car
  - Basic driving techniques
  - Safe driving skills
  - Looking out for hazards
  - Getting though roundabouts
  - Safe stopping distances.

Off-road driving

Where there is a need for off-road driving the Tour Operator will usually provide a four-wheel drive vehicle to handle these conditions.

Pre-requisites

Before driving off-road it is advisable to (where possible/practicable):

- Ensure the vehicle has been properly prepared for the conditions which are known to exist or which are expected on the trip – this may include:
  - Loading the vehicle with necessary recovery gear
  - Fitting protection and bash plates to the underside of the vehicle
  - Fitting appropriate tyres
  - Having tyre deflators and pumps
  - Having warning/sand flags
  - Having chains
  - Fitting a snorkel
  - Fitting a viscous fan
  - Fitting extended differential breathers
- Undertake basic checks – to confirm:
  - The vehicle is fully operational – to the nest extent possible under the circumstances that apply at the time
  - Safety and recovery gear is available and operational
- Inspect or consider the way ahead – to:
  - Determine the route/path to be taken
  - Identify potential obstacles/problems
  - Make the final 'Go' or 'No go' decision.

- Always be prepared to, as required:
  - Choose or find an alternate route
  - Back-track and go through using another track
  - Wait for poor conditions (high water levels, high flow rates of rivers, deep snow) to abate/improve before proceeding
- Consider passenger safety – which may involve:
  - Communicating with them to advise them of what to expect
  - Providing safety information
  - Keeping them involved and included in the tour experience

  Taking **controlled** risks can be part of the excitement and experience of a 4WD adventure but they must never be undertaken where they pose an uncontrolled threat to people/passengers.
- Engage four-wheel drive – as appropriate for the individual vehicle and the nature of the route and terrain to be driven.

Driving on sand

Preparing for sand

When preparing for soft sand:

- Lower tyre pressure of 4WD vehicle – as a starting point to increase the grip area of the tread and help keep it on top of the sand instead of sinking (or biting) into it:
  - Reduce by half standard on-road pressure
  - Reduce more for softer surfaces
- Lower tyre pressure of trailer, where appropriate
- Use pressure gauge to determine pressures – so there is a known reference point: write the pressures down as they are obtained and trialled
- Be prepared to lower pressures further if first decrease appears insufficient – reduce by approx. two to four psi (15 – 30 kPa) each time.

Read more at:

http://www.offroadaussie.com/2013/02/4wd-tips-and-tricks-tyre-pressure/ - 4WD tricks and tips: tyre pressure.

Driving on hard sand

Driving on compacted sand is similar to driving on sealed roads in many ways.

Techniques for driving a 4WD on hard sand include:

- Reduce tyre pressure slightly – to increase surface area of tread that makes contact with the sand
- Engage 4WD – which may require locking in hubs
- Select High range
- Drive off choosing gear/s appropriate to conditions, weight of vehicle, route being used and desired speed of travel
- Travel at a reduced speed – that is, slower than on the open highway
- Avoid contact with salt water wherever possible – to prevent potential of vehicle being damaged by rust: always wash any vehicle as soon as possible that has been contacted by salt water
- Re-inflate tyres when sand driving is completed – if they have been tyre pressure was initially reduced.

Driving on soft sand

Indicative actions/techniques are:

- Deflate tyres – to around 16 psi/110kPa or lower depending on sand depth/condition
- Attach sand/warning flags to the vehicle if driving in areas where there are hills/dunes – to hep indicate location of the vehicle to others in the area
- Check recovery gear is available – to use should the vehicle become stuck
- Engage 4WD – which may require locking in hubs
- Select Low range – second gear is a good starting point
- Start slowly – to avoid wheel spin and digging in
- Drive slowly but maintain momentum
- Avoid sudden turns
- Drive in established tracks – where the sand will be more compacted by the vehicles which have previously used that route
- Be prepared for less feedback from the steering wheel – due to the depth of the sand
- Approach any dunes head on/square on if possible – to avoid potential for roll-over
- Stay alert for oncoming traffic over the top of dunes
- Never assume the drop off on the other side of a dune is the same as the climb up the dune – drop-offs on the other side can be substantially different due to wind action which may have created a sheer fall/drop away on the other side
- Use the brake gently – to avoid sand building up in front of the tyres
- Be prepared to reverse back over the tracks made by the vehicle if stuck – as these will be more compacted and may provide sufficient traction to develop the required traction and momentum to move forward
- Re-inflate tyres when sand driving is completed – if they have been tyre pressure was initially reduced.

Websites

See more information at:

- http://4xoverland.com/how-do-4x4s-work/4x4-driving/ – How to drive a 4x4 off road [Sand driving]
- https://www.youtube.com/watch?v=iT0Zph4h-L4 – ExpeditionOz 4WD Skills: Sand driving
- https://www.youtube.com/watch?v=I2bA6aC4g50 – Soft sand driving
- https://www.youtube.com/watch?v=KHI_cMrvsrM – 4x4: How to drive on sand tracks
- http://www.outbackcrossing.com.au/FourWheelDrive/How_To_Drive_0n_Sand.shtml - How to drive on sand
- https://www.youtube.com/watch?v=Int_D9zKjiY – Tips for beginners: Driving on sand, dunes and beaches
- http://www.iventureclub.com.au/4x4-tips/article.html?k=YqKAOl33YuQ – 4x4 tip: Sand driving.

Driving on very soft ground or through mud

Techniques/indicative action includes:

- Inspecting the route – looking for:
  - Ruts and obstacles
  - Condition of the bottom under the surface mud/depth of the mud
  - Distance to be covered
  - Recovery points
- Dropping tyre pressure – to around 24 psi/165kPa
- Engaging 4WD
- Locking diffs
- Selecting Low range – first or second gear
- Entering the mud at a slow to moderate speed – avoiding excessive speed
- Using or not using ruts as appropriate – sometimes it is best to drive in the ruts and sometimes it is best to avoid them, for example:
  - Driving in ruts (which can provide a more compacted surface and therefore increase traction) if there is sufficient clearance for the bottom of the vehicle
  - Straddling the ruts – where the *adjacent* ground is firmer and will give better traction
- Maintaining momentum – by applying acceleration
- Spinning tyres – where mud tyres are fitted to help clean the tread
- Being prepare to stop, reverse and re-try a route if progress is halted – as this may give extra traction through the more compacted bottom surface
- Moving the steering wheel left-to-right – to allow side lugs of tyres to bite the mud.

Website

See more information at:

- http://4xoverland.com/how-do-4x4s-work/4x4-driving/– How to drive a 4x4 off road [Mud driving]
- https://www.youtube.com/watch?v=O7iqHO8__R8 – Mud driving the right and wrong way
- http://www.iventureclub.com.au/4x4-tips/article.html?k=hmcpHxSX7pc – 4x4 Tip: mud driving
- http://www.toughtoys.com.au/how-to-4x4/mud-driving-techniques/ – Mud driving techniques.

Driving across rocky terrain

In general, rocky terrain should be avoided wherever possible as it poses lots of risks in terms of:

- Uncomfortable ride for passengers
- Slow travel times – which may negatively impact itinerary
- Punctures
- Damage to the vehicle especially the chassis and sub-structure/under-side parts of the vehicle such as diffs and fuel tanks.

When driving rocky terrain:

- Inspect the path – as described above
- Reduce pressures in tyres
- Engage 4WD – which may require locking in hubs
- Engage diff locks – if fitted and if necessary
- Turn off air conditioner – to avoid inconsistent engine acceleration
- Begin in a manner that reflects the findings of observation – which may require, as appropriate:
  - Selecting Low range first or second gear
  - Driving slowly ('crawling') and smoothly to avoid wheel spin
  - Intelligently choosing the path to take to avoid traps, rocks and beaching of the vehicle
- Be prepared to ask passengers to get out of the vehicle – if patches of the route are dangerous
- Be prepared to ask passengers to act as spotters – to assist with providing directions on where to position wheels
- Try to stay on the high parts – straddling any ruts which may be present
- Re-inflate tyres when the rocky terrain has been negotiated.

Websites

Check out the following:

- http://www.4-wheeling-in-western-australia.com/rock-crawling.html – Rock crawling and rocky obstacles (including viewing of all videos linked to this site)
- http://www.iventureclub.com.au/4x4-tips/article.html?k=1ImWwpAqbd0 – 4x4 tip: Rocks and ruts
- http://www.toughtoys.com.au/how-to-4x4/driving-rocky-terrain/ – Driving rocky terrain.

Driving inclines

Ascending steep inclines

Indicative actions are:

- Inspect the path – looking for:
    - A suitable route through/across the track which presents itself – taking into account, for example, ruts, holes, fallen trees, rocks
    - Surface condition – in relation to whether the ground is wet or dry and whether it is hard underfoot or soft
    - Approach angle – to ensure the vehicle can begin the climb and get its wheels onto the incline, and not get 'beached'/stuck trying to make the climb
    - General angle/severity of the climb
    - Length of the incline – in terms of the distance to be covered to compete the climb
    - Possible recovery points that can be used – if the vehicle becomes stuck

- Engage 4WD – which may require locking in hubs
- Engage diff locks – if fitted and if necessary
- Turn off air conditioner – to avoid inconsistent engine acceleration
- Begin the climb in a manner that reflects the findings of observation of the incline – which may require, as appropriate:
    - Reversing and taking a run at the incline to gain momentum perhaps in H4 where the ground is solid, the length of the climb is short and the angle of incline is not too steep
    - Selecting Low range first or second gear and driving slowly up the incline where it is a steep incline, ground is soft sand/or the distance to be covered is significant
- Stay square to the incline – climbing at an angle to the rise introduces the potential for the vehicle to roll
- Work the wheel a little from side-to-side if the wheels start spinning – in order to use the tread on the sides of the tyres to gain additional purchase/traction
- Pause at the top of the incline – in order to:
    - Check the descent – if there is one (see below)
    - Check for oncoming vehicles.

Stall recovery

If the vehicle stalls when climbing a steep incline the procedure for a safe stall recovery is:

- Apply foot brake as vehicle stalls
- Apply handbrake
- Turn ignition off
- Depress clutch
- Select reverse gear
- Release clutch
- Release handbrake slowly
- Turn on ignition and release brakes
- Start vehicle allowing it to reverse down the slope
- Apply brakes as necessary to control descent.

Descending steep inclines

Standard advice regarding steep descents is to avoid them where possible and find an alternate route especially where the route is wet or otherwise presents an extra risk on top of the angle of descent.

Indicative actions may include:

- Inspect the path – looking for:
  - A suitable route through/across the track which presents itself – taking into account, for example, ruts, holes, fallen trees, rocks

  - Surface condition – in relation to whether the ground is wet or dry and whether it is hard underfoot or soft
  - Departure angle at the end of the incline – to ensure the vehicle can leave the incline and make it to level ground instead of being stuck half-way between the incline and the flat ground
  - Length of the descent – in terms of the distance to be covered to compete the descent
  - General angle/severity of the descent
  - Possible run-off points (into banks, bushes, trees) – in case the descent becomes out-of-control
- Engage 4WD – which may require locking in hubs
- Engage diff lock
- Turn off air conditioner – to avoid inconsistent engine acceleration
- Begin travel in a manner that reflects the findings of observation of the descent – which may include:
  - Selecting Low range first gear
  - Allowing vehicle to creep forward under its own power without using the accelerator – letting the engine act as a brake

- Applying extra braking as required using either foot brake or hand brake – to keep a safe speed
- Minimising use of brake – allowing engine to do the braking
- Avoiding locking of wheels – the wheels must continue to rotate in order to retain control/steering
- Keeping wheels pointing forward
- Stay square on to the incline – to avoid potential for roll-over.

Stall recovery

If the vehicle stalls when descending a steep incline the procedure for a safe stall recovery is:

- Turn ignition off
- Apply foot brake
- Engage handbrake
- Depress clutch
- Select low gear
- Release clutch
- Slowly release handbrake
- Turn ignition on
- Release brakes
- Start vehicle and allow to continue down the slope
- Apply brakes emulating ABS as necessary to control descent
- Steer into skids.

Websites

See more information at:

- http://www.4x4australia.com.au/gear/1004/4x4-steep-slopes-tips/ - 4x4 steep slope tips
- http://4xoverland.com/how-do-4x4s-work/4x4-driving/ - How to drive a 4x4 off road [Steep slopes]
- https://www.youtube.com/watch?v=uTW3H4WBxnY – 4WD Hill Climb Comparo
- http://www.iventureclub.com.au/4x4-tips/article.html?k=Jem4UGra2DY – 4x4: Tip- Steep terrain.

Driving through water

Preparing for water

Standard requirements before making deep-water crossings are:

- Know the vehicle and how deep water the water can be before the vehicle begins to float
- Know how fast the water is flowing – to determine safety and work out speed required and angle of attack
- Know the surface to be travelled – on the bottom of the river and on exit from the crossing
- Know where the air intake is – to make sure the vehicle does not ingest water
- Check-out the crossing – walk through the crossing to determine depth, identify obstacles and work out the path to be taken:
  - Walk across the river where the left-hand wheel will go
  - Walk back where the right-hand wheel will go
- If in doubt, avoid the crossing.

Standard cautions

Be prepared to avoid the crossing if:

- The current is too fast
- The water is too deep
- The approach or departure angles are too steep
- The approach or departure are too soft or otherwise too difficult
- The vehicle is not/cannot be adequately prepared.

Driving techniques

When driving through water there can be a need to:

- Let the engine cool down first – the steps below will assist with this
- Tie a tarpaulin across the front of the vehicle – to help create a bow wave
- Loosen the fan belt – to prevent damage to the radiator (unless vehicle is fitted with a viscous fan)
- Spray the motor with water repellent (WD-40) to help preserve the electrics
- Attach recovery straps to the vehicle – in case there is a need to recover the vehicle from the middle of the river: make sure these straps are secured above the water line
- Lower windows – as a safety factor
- Advise passengers not to wear seat belts
- Lower pressure in tyres – if exit is sandy/soft
- Select 4WD, low range – first or second gear depending on type and size of engine
- Lock the diffs

- Enter the water at a slow to moderate speed – so as to create a bow wave but not a ‘splash’ which will throw water up and over the bonnet/vehicle
- Drive slowly and evenly
- Avoid using the clutch – as water will enter the housing and cause damage, slippage
- Dry brakes after exiting the water.

Website

Take the time to visit:

- http://www.iventureclub.com.au/4x4-tips/article.html?k=zFA1eN-qEzg – 4x4 Tip: Water crossing
- http://www.toughtoys.com.au/how-to-4x4/water-crossings/– Water crossings
- https://www.youtube.com/watch?v=wNtJNuhIf4I – ARB Water crossing techniques
- http://www.outbacktravelaustralia.com.au/driving-towing-4wd-driving-skills/water-crossings – Water crossings
- http://www.4-wheeling-in-western-australia.com/river-crossings.html– River crossings [including videos on this page]
- http://www.4wdonline.com/A/Water.html – Water crossing
- http://www.pps.net.au/4wdencounter/4wdtech/water.html – 4WD Encounter Australia: River crossings.

4.2 Demonstrate practices related to the towing of a trailer

Introduction

Drivers of service vehicles are often required to tow a trailer to carry camping gear, food and other equipment to support a tour group or when there is a need to transport groups with large amounts of luggage.

This section discusses considerations related to towing practices.

Some countries:

- Only allow fully-qualified drivers (holders of 'full' driving licences) to drive a vehicle that tows a trailer
- Require trailers to be registered and roadworthy before they can be used on the road.

Loading of the trailer

When stowing gear on a trailer the following may apply:

- Use a checklist to ensure all necessary items are loaded
- Balance the weight of the load to the best extent possible based on the size and weight of individual items – to optimise stability: balance should be approximately 60% towards the front of the trailer and 40% to the rear
- Secure items to prevent movement during travel
- Load gear/items required first, last – so they are closer to the door when the trailer is opened
- Place heavier items on floor of trailer – or 'lower down'
- Use cargo nets to help control/limit movement of luggage
- Use ropes to tie down camping gear and equipment.

Attaching the trailer

When attaching the trailer indicative actions are:

- Ensure correct size tow ball – to match tow coupling/tow ball weight
- Ensure correctly rated D-shackles, bolts and safety chains
- Connect trailer using jockey wheel to help final positioning of trailer and lowering onto tow ball
- Tighten hand wheel – to secure ball clamp
- Hook up connections as required – including:
  - Lowering trailer coupling over tow ball using jockey wheel
  - Ensuring coupling is correctly positioned/seated
  - Connecting and tightening chains and shackles
  - Attaching electrics for lights

    - Attaching brakes
    - Attaching load distribution bars and/or sway controls – where fitted/available
- Raise and secure jockey wheel.

Towing a trailer

Tips when towing a trailer include:

- Remember there is a trailer connected to the vehicle – do not forget it is there
- Slow down – travel at a reduced speed when towing a trailer
- Allow extra space between the vehicle and the traffic in front – to allow for the extra braking distance required to accommodate the extra weight and length of the trailer
- Observe the trailer on a regular basis through the mirrors – to detect any problems at the earliest opportunity
- Check the trailer at all stops – as part of standard vehicle inspection protocols
- If the trailer starts to sway, take foot off the accelerator – do not hit the brake and do not try to accelerate out of this kind of trouble
- Factor in extra time when overtaking – to allow for the added length of the trailer
- Do not travel in the 'fast' lane
- Make sure there is enough room for the vehicle *and trailer* to get into any space being considered
- Realise the extra time/space required due to the extra length of the unit when entering a road and getting up-to-speed
- Take corners a little wider – to allow room for the trailer to properly negotiate the corner
- Fit extension mirrors – to try to reduce extra blind spots created by the trailer.

Detaching the trailer from the towing vehicle

When detaching the trailer, indicative actions include:

- Make sure vehicle is parked on flat ground – to avoid potential for trailer to run away
- Apply hand-brake to vehicle – and put vehicle in gear or 'Park'
- Apply trailer brake
- Unload trailer before disconnecting – if practicable
- Detach and secure safety chains
- Disconnect and secure electrics, brakes/air lines, load distribution bars and/or sway controls – as relevant
- Drop jockey wheel
- Undo hand wheel – to take pressure off ball clamp
- Use jockey wheel to raise coupling from tow ball.

Practical driving

Practical aspects of this unit must include:

- Towing a trailer up-hill
- Towing a trailer down a hill
- Towing a trailer across level ground
- Towing a trailer across/through water.

4.3 Demonstrate reversing practices

Introduction

Standard road and travel conditions will mean there can be a need to reverse the vehicle and this is also a mandatory part of the driving test.

This section presents advice to follow when reversing a vehicle and a trailer.

Keys when reversing

Important points to note when reversing are:

- Do not rush
- Check the area and/or path to be reversed into – never assume it is clear and/or safe to use
- Use the mirrors
- In some countries it is illegal to reverse on freeways – because it is deemed too dangerous
- Limit the amount/distance of reversing that is done.

Websites

Take time to view:

- http://www.qld.gov.au/transport/licensing/getting/education/videos/ – Safe driving videos: 'The reverse park'
- https://www.youtube.com/watch?v=_FzNRQFhSSk – How to reverse your car correctly
- https://www.youtube.com/watch?v=orDT9Sm_Y1I – Driving 101: how to reverse into a parking space
- https://www.youtube.com/watch?v=TOMUo7LXE68 – How to drive a car in reverse gear easily.

Reversing a trailer

Reversing a trailer can be difficult for many.

It takes lots of practice to become proficient.

Keys are:

- Take it slowly
- Use the mirrors
- Turn the steering wheel in the **opposite direction** to which the trailer is required to move
- Realise the **direction of the bottom of the steering wheel** is the direction in which the trailer will move.

Websites

https://www.youtube.com/watch?v=cqW73F8CbXM – How to reverse a trailer

https://www.youtube.com/watch?v=jo0MLjY8ppU – Towing: how to back-up and park a trailer.

Practical driving

Practical aspects of this unit must include reversing a trailer:

- Up-hill
- Down-hill
- On level ground
- Into confined spaces.

4.4 Demonstrate parking and shutting down procedures

Introduction

All vehicles need to be shut down and parked when they arrive at destinations and/or return back to base.

This section provides advice on parking and shutting down the vehicle.

Parking and shutting down

Exact parking and shutting down procedures will:

- Vary between different makes and models of vehicles
- Differ between Tour Operators according to their individual SOPs.

Points to note when parking and shutting down are:

- Locate the vehicle in a safe position – so that:
  - Its physical security is maintained as best as possible
  - It does not create a risk/danger in its own right by virtue of where it is parked – by (for example):
    - Blocking line of sight of other drivers to important aspects of the road
    - Blocking access or impeding the flow of traffic
- Park legally – which may mean:
  - Parking in designated parking areas/space
  - Parking 'between the lines'

- Seek to drive into a parking area – rather than reverse into one
- Give other vehicles 'sufficient' space and room to manoeuvre
- Apply the parking brake when stopped
- Leave the vehicle in gear or 'Park'
- Shut down the vehicle – by turning off accessories and turning the engine off
- Illuminate parking lights if leaving the vehicle on street/road at night
- When parked on an incline, turn the front wheels in to the kerb
- Secure valuable items out-of-sight – to help prevent break-ins
- Activate/apply anti-theft devices – if leaving the vehicle
- Secure the vehicle when leaving it – SOP is to lock the vehicle
- Complying with any other manufacturer's instructions which apply – to the individual type, make and model of vehicle
- Adhere to all Tour Operator standard operating procedures – as these apply to parking and/or shutting down the vehicle.

Websites

- https://www.youtube.com/watch?v=PtygeD4t6ig – How to park a car
- https://www.youtube.com/watch?v=199QsWyBhpo – Parallel parking lesson
- https://www.youtube.com/watch?v=-8AAciq3wqo – Bay parking lesson: full version
- https://www.youtube.com/watch?v=BF44oWTY47M – Bay parking on driving lessons.

4.5 Identify and modify driving techniques to accommodate driving hazards

Introduction

It is to be expected all drivers will adjust their driving to accommodate hazards and match emerging traffic and road conditions.

This section provides more advice on safe driving.

Identification of hazards

Driving hazards which may be encountered can include:

- Wet and iced roads
- Oil on the road
- Animals and objects on the road
- Parked vehicles on the road
- Pedestrians on and crossing the road
- Flooded roads
- Windy sections of a road
- Reduced visibility.

Generic techniques to address hazards

Generic requirements for coping effectively with these conditions are:

- Constant vigilance when driving – especially in relation to observational skills
- Talking to other drivers on the road to obtain up-dates and advice on local conditions
- Listening to two-way radio chatter regarding road and traffic conditions
- Advising others about hazardous conditions encountered – when and if safe to do so
- Slowing down
- Using common sense
- Never placing passengers or the vehicle at risk
- Abiding by all applicable road laws.

Specific techniques

Wet roads

- Slow down
- Turn headlights on
- Activate wipers
- Activate de-misters
- Allow greater distance to vehicles in front – to give more braking distance and avoid spray
- Exercise greater patience
- Realise the road surface is most slippy when the rain starts – as a result of oil/grease build-up which has not yet been washed away
- Allow more travel time for distances to be covered.

Iced roads

A wet looking road but no spray coming from the tyres indicates driving on ice.

- Consider delaying travel – which may mean stopping and waiting for improved conditions
- Consider an alternate route
- Notify Tour Operator of problem – and any action taken which needs to be reported
- Leave greater braking distances
- Turn headlights on
- Apply brakes gradually – to avoid skidding
- Move the steering wheel gently
- Activate heaters and de-misters
- Realise ice forms more readily on bridges (due to lack of sub-surface heat) – so be extra careful on these in cold/icy conditions.

Oil on the road

- Try to avoid the oily patch
- Slow down
- Avoid heavy braking in the oil
- Notify Tour Operator of problem – so other drivers can be advised.

Animals and objects on the road

- Slow down
- Sound horn
- Try to avoid the animal or object if possible
- It is better to hit an animal or object than it is to hit another vehicle or a pedestrian
- Driving at night increases chance of hitting animals
- Dawn and dusk are the worst times for hitting animals on the road
- Stop to inspect vehicle if contact is made.

Parked vehicles on the road

- Slow down
- Sound horn
- Turn on headlights
- Ensure sufficient room to pass
- Get out to check/measure distance/gap if necessary
- Travel very slowly where room is tight
- Be alert to people opening doors into the path of the vehicle
- In tight situations, use a spotter to assist
- Use mirrors to check progress
- Consider waiting until vehicle is moved
- Consider alternate route.

Pedestrians on and crossing the road

- Always be alert to the presence of foot traffic
- Pay extra/special attention to children as they have no road knowledge and no understanding of the dangers involved in running out onto the road
- Drive with headlights on
- Sound horn
- Slow down
- Give those walking on the road a wide berth
- Pedestrians on crossings have right-of-way
- Never assume pedestrians have seen the vehicle/know it is there
- Never assume pedestrians will wait.

Flooded roads

- Follow Tour Operator SOPs regarding crossing flooded roads
- Err on the side of caution
- There is generally a ban on driving a vehicle through flood water on roads
- Never put the safety of the passengers or the vehicle at risk
- Turn headlights on
- Advise passengers to unfasten their seat belts
- Use low gear
- Seek alternate route
- Delay travel until water subsides
- Notify Tour Operator of problem – and any action taken which needs to be reported.

Windy sections of a road

- Slow down
- Realise the side of the vehicle will act like a sail
- Steer slightly into the wind
- If practical take a line towards the side of the road from which the wind is blowing
- Seek an alternate route
- Delay travel until wind drops to an acceptable level
- Notify Tour Operator of problem – and any action taken which needs to be reported and/or so other drivers can be advised.

Reduced visibility

- Stop and delay travel until visibility improves
- Turn on headlights
- Activate emergency flashers
- Sound horn
- Reduce speed according to distance that can be seen
- Realise other drivers will be experiencing the same
- Seek an alternate route
- Notify Tour Operator of problem – and any action taken which needs to be reported.

4.6 Modify driving style to accommodate identified vehicle problems and faults

Introduction

From time-to-time there can be a need to modify normal driving style in order to take into account problems which have emerged with the vehicle.

This section discusses responses in this regard.

Standard responses

The basic responses to driving a vehicle with an identified fault include:

- Adherence to company policies as they apply to various situations
- Never put the safety of passengers or other road users at risk
- Use common sense
- Remember the Duty of Care owed by drivers to others – see below
- Notify the Tour Operator and advise of the problem – and seek their direction about what to do.

Duty of Care

Common law imposes an obligation on all vehicle drivers to:

- Create and maintain a touring environment that does not pose a risk to people (staff, members of the public, customers)
- Take action to avoid causing foreseeable harm to people/customers/tour group members or their property/belongings while they are on tour
- Avoid engaging in risky behaviour or actions which will pose a threat to anyone.

In practice this means the driver needs to/has responsibility for ensuring:

- The vehicle is safe/roadworthy when it departs
- The safety/roadworthiness of the vehicle is maintained through the trip
- All road rules and speed limits are obeyed
- The vehicle is driven in a safe and responsible manner
- Obvious risks are avoided or controlled
- Passengers are informed about potential risks/hazards
- Passengers comply with safety rules
- Cargo is stored/stowed appropriately – which may require:
  - Re-packing/re-locating items for off-road 4-wheel drive travel to attain a lower centre of gravity for the vehicle
  - Placing items behind physical protection barriers – or otherwise securing them to prevent injuring people or causing vehicle damage.

Consequences of failing to meet Duty of Care obligations

If drivers fail to meet their Duty of Care obligations:

- They may be sued by passengers/those they injure or who lose property for negligence
- The Tour Operator they work may be sued for not discharging their Due Diligence obligations by properly training their drivers.

The outcomes of successful civil claims for negligence are commonly the awarding of money ('damages') to the person who has been injured or suffered loss.

Important notes

It is extremely important for drivers to understand:

- Many problems are a matter of degree – as opposed to 'total failure', so it is often possible to keep going **provided it is safe to do so** given the situation and the circumstances that apply at the time
- Passengers will be extremely uncomfortable and often very critical if they know there is a problem with the vehicle and it is not fixed
- Severe problems relating to brakes, steering or suspension mean there is a need for attention to the issue as opposed to a modification of driving style
- Problems relating to vehicle performance (acceleration, rough running, minor suspension issues) may be able to persevered with in the short-term in order to reach a known service point and/or to get passengers to a destination (more or less) on time
- Problems with over-heating *may* be able to be addressed by travelling more slowly, more regular stopping to allow the motor to cool down, or effecting roadside repairs (such as replacing fan belts or radiator hoses)
- Loss of lights at night means the vehicle must not be driven – until suitable repairs are made
- Standard practice when a problem emerges is to:
  - Do nothing to risk causing further damage to the vehicle or to make matters worse
  - Slow down, and continue to drive while continuing to monitor the situation using:
    - The senses and personal observation
    - The gauges on the vehicle
  - Stop, where the problem has been identified as significant – and wait for service/mechanics to attend and fix the problem
  - Be prepared to call for a replacement vehicle – and move passengers and their luggage/belongings to another vehicle.

4.7 Follow emergency procedures

Introduction

Drivers are expected to follow pre-prepared emergency procedures in the event they encounter an emergency when driving.

This section gives a general context for responding to emergency situations and give suggestions for action in a number of common.

Context

All Tour Operators will prepare procedures for their drivers to follow in the event they need to respond to a range of emergencies.

These responses/procedures will be included as part of the on-the-job training provided to new drivers.

It is a non-negotiable requirement:

- All organisational protocols must be adhered to in these circumstances – subject to exercising common sense and good judgement in every situation given the exact nature and scope of the emergency
- All legal obligations must be discharged
- The safety of passengers is paramount at all times
- The safety of people (other road users and members of the public) must always take precedence over the security/saving of assets.

Dangerous vehicle emergencies

The following based on information from 'The Victorian Bus and Truck Drivers' Handbook' (https://www.vicroads.vic.gov.au/searchresultpage?q=bus%20inspection) which offers the following at pp. 72 – 77:

> **"Dangerous vehicle emergencies**
>
> There are three vehicle emergencies that can be very dangerous:
>
> » brake failure
>
> » tyre failure
>
> » fires
>
> ***Brake failure***
>
> Brakes kept in good condition rarely fail. Most brake failures occur for the following reasons:
>
> » brake fade on long hills
>
> » poorly maintained brakes

Here are some actions you can take:

» change **down gears.** If you are able to change down gears, this will help to slow the vehicle

» pump **the brakes.** Sometimes pumping the brake pedal will produce enough hydraulic pressure to stop the vehicle

» use **the parking brake.** The parking brake is separate from the hydraulic brake system, so it can be used to slow the vehicle. Be sure to press the ratchet release button (if fitted) and keep it pressed at the same time you use the brake. You can then adjust the brake pressure and stop the wheels from locking

» find **an escape route.** While slowing down, try to find an escape route. It could be a driveway, an open paddock, or a side street. In larger hydraulic brake vehicles, the emergency braking system will start working when brake fluid problems develop. Be prepared for a skid when this happens.

<u>Brake fade on long hills</u>

Being in the proper gear and braking properly will prevent most brake failures on long hills. But if the brakes have failed, you will have to look outside your vehicle for something to stop it.

If there's a safety ramp, road signs will tell you about it. Use it.

If there is no safety ramp, take the best escape route you can, such as an open paddock, or a side road that flattens out or turns uphill. Make the move as soon as you cannot control the vehicle. The longer you wait, the faster you'll go, so it will be harder to stop.

Tyre failure

Tyre failure on one of the drive wheels or trailer wheels will not usually cause a crash. Failure of one of the front tyres could cause a loss of steering control.

To avoid tyre failure check your tyre regularly during your trip.

There are four important things that safe drivers do to handle a tyre failure.

» know the signs that a tyre has failed (see below)

» grip the steering wheel firmly, with both hands

» stay off the brake

» apply the trailer brake lightly if a steering tyre blows.

<u>Recognise failure signs</u>

If you know that you have a tyre failure, you can do the right thing and do it quickly. The main signs of tyre failure are:

» sound**.** Although many tyre failures cannot be heard, the loud "bang" of a blow-out is easily recognised.

Because a few seconds elapse before the vehicle reacts, many drivers at first assume it must have been another vehicle. So any time you hear a tyre blow, you should assume it is yours:

» vibration**.** If the vehicle thumps or vibrates heavily, it may be a sign that one of the tyres has gone flat. With a rear tyre, this may be the only sign you get

» feel**.** If the steering feels heavy, it is probably a sign that one of the front tyres has failed

Any of these signs should be a warning of a possible tyre failure.

If your tyres have failed, you should:

» grip **the wheel firmly.** When a front tyre fails, it can twist around the rim, exerting such a powerful force that it could snatch the steering wheel out of your hands. The only way to stop this happening is to have a firm grip on the steering wheel with both hands.

Keep your thumbs out from under the spokes of the wheel. Your thumbs could get broken if the steering wheel snaps around before you can get control of it.

» stay **off the brake.** It is natural to want to brake in an emergency. However, in a tyre failure, it could make the wheels lock up and result in a skid. Unless you are about to run into something, stay off the brake until the vehicle has slowed down. After you've slowed down, then brake very gently, pull off the road, and stop

» check **the tyres.** After stopping, get out and check all the tyres. Do this every time you stop even if the vehicle seems to be handling correctly. If one of your dual tyres deflates, the only way you may know, is by getting out and checking it.

Fires

Vehicle fires are a frequent cause of damage and injury. Learn the cause of fires and how to prevent them.

Know how to extinguish fires.

Causes of fire

The major causes of large vehicle fires are listed below. All of these fire causes can be avoided!

» crashes**.** Spilled fuel after a crash

» tyres**.** Under-inflated tyres and dual tyres that touch each other

» brakes**.** "Riding" the brakes or excessive use of brakes on hills causes linings to over-heat and ignite the wheel lubricant

» wheel **bearings.** Running hot (not enough lubricant)

» electrical **system.** Damaged insulation, loose wires

» exhaust **system.** Lack of proper insulation, parking in dry grass, sparks or hot exhaust gas

» fuel**.** Driver smoking, improper fuelling, loose fuel connections

» load [cargo/luggage]**.** Flammable loads, improperly sealed or loaded, poor ventilation.

<u>Fire fighting</u>

Pull off the road. The first step is to get the vehicle off the road and stop.

In doing so:

» park in an open area away from buildings, trees, bushes, other vehicles or anything that might catch fire

» do not pull into a service station.

Stop immediately. In the event of a tyre fire, stop immediately.

Notify emergency services. Use the nearest public telephone or, if available, your mobile telephone or CB radio to notify the Police and Fire Brigade of your problem and your location. They can often get fire-fighting equipment to you.

When describing your location, give as much information as possible:

» name of street

» suburb

» in country areas:

› distance from the nearest town

› a landmark that will help identify where you are.

Stop the engine. Isolate battery or batteries if possible.

Stop the fire from spreading if it is safe to do so. Otherwise, ensure that any members of the public have been moved to a safe distance.

Before trying to put out the fire, make sure that it does not spread any further.

With an engine fire, keep the bonnet or engine cover closed. This stops air reaching the fire to fuel it. If you can, shoot a dry powder extinguisher or foam through louvers, radiator grill or from underneath the vehicle.

For a load fire in a van or box trailer, keep doors closed until you have sufficient extinguishers and help available, or leave closed and notify the Fire Brigade.

When a tyre is smouldering and cannot be extinguished try to remove it. Get it away from the vehicle, only if safe to do so.

To extinguish a tyre fire, water is the best extinguishing agent. A large dry chemical extinguisher may present the next best extinguishing medium. In all cases the Fire Brigade must be notified as soon as possible.

If a trailer fire is out of control and it is safe to do so, quickly unhook the prime mover and get it away from the fire.

<u>Use the correct extinguisher</u>

Using the wrong type of extinguisher could spread the fire and make it worse.

Know the type of extinguisher that is in your vehicle. Make sure you’ve read the extinguisher’s instructions before driving. It is hard to read the fine print during an emergency.

On an electrical fire, do not use water (water conducts electricity).

On a fuel fire, do not use water (it will just spread the flames). For best results, get trained in extinguisher use.

Putting out the fire

Here are some rules to follow when you have to put out a fire.

» know how the fire extinguisher works. Every time you see a different type of extinguisher, find out how it works and, if possible, get practical experience in using it

» stay away from the fire. Use the full squirting distance of the extinguisher

» aim at the source or base of the fire, not up in the flames

» position yourself upwind. Let the wind carry the extinguisher contents to the fire rather than carry the flames back to you

» continue using the extinguisher until whatever was burning is cool. No smoke or flame does not mean the fire is out or will not start again.

Vehicle breakdown

If the vehicle breaks down while being driven, standard aims and actions are to:

- Pull to a safe position – off the road to the best extent possible
- Activate hazard warning lights
- At night, make sure side/running lights and parking lights are left on
- Strive to position the vehicle in/under protection from the elements (sun) – to give protection to passengers
- Immediately notify the Tour Operator/Head Office – advising them of:

  - Location
  - Nature of the problem to the best extent known at this early stage
  - Proposed course of action
  - Need for direction/advice in relation to addressing the issue
  - Urgency which may be associated with the breakdown especially if the vehicle has come to rest in a dangerous position which poses a risk to:
    - On-board passengers
    - Other road users
    - Members of the public
- Deploy warning triangles in advance of the broken down vehicle – to warn on-coming vehicles.

In the event of a collision

'The Victorian Bus and Truck Drivers' Handbook' (https://www.vicroads.vic.gov.au/searchresultpage?q=bus%20inspection) pp. 80 – 81 gives the following advice:

> "**What you must do at a crash**
>
> When you are involved in a crash or come upon a crash you need to take prompt and proper action to prevent further injury or damage.
>
> The basic steps to be taken at any crash scene are:
>
> » stop
>
> » protect the area
>
> » care for injured
>
> » notify authorities - where are you? How many injured?
>
> » provide your name, address and registration details and the vehicle owner's name to the other parties involved in the crash, or their representative, and to the police (if in attendance).
>
> It is a serious offence if you fail to:
>
> » stop and help at a crash
>
> » exchange all required details after being involved in a crash.
>
> ***Protect the area***
>
> The first thing to do at a crash scene is to prevent another crash from happening at the same spot.
>
> The crash area needs to be protected.
>
> If your vehicle is involved in a crash, try to get it to the side of the road to prevent another crash and to allow traffic to move. Use hazard warning lights or your portable warning triangles.
>
> If you are stopping to help, park away from the crash. Emergency vehicles will need to get close. Put on your hazard warning lights. A large vehicle is easier to see than a car and its hazard warning lights will be more noticeable.
>
> Set out portable warning triangles or emergency lamps to warn other traffic. Make sure they can be seen.
>
> **Notify authorities**
>
> If you have a CB radio, put out a call over the emergency channel before you get out of the vehicle. Do not forget to say exactly where you are. Otherwise, wait until after the crash scene has been properly protected and then send someone to phone the police. Tell the police if anyone has been hurt and, if so, how many people are injured. Try to determine exactly where you are so you can give the exact location. A long and sometimes fatal delay can occur if rescue people are sent to the wrong place."

Work Projects

It is a requirement of this Unit you complete Work Projects as advised by your Trainer. You must submit documentation, suitable evidence or other relevant proof of completion of the project to your Trainer by the agreed date.

4.1 To fulfil the requirements of this Work Project you are required to drive at least two different vehicles (to be approved by your Assessor) providing evidence you have:

- Demonstrated basic operational driving practices as required by the Assessor
- Demonstrated practices related to the towing of a trailer as required by the Assessor
- Demonstrated reversing practices as required by the Assessor for both vehicles
- Demonstrated standard parking and shutting down procedures for the as required by the Assessor for both vehicles
- Identified and modified driving techniques to accommodate driving hazards as presented by the Assessor for both vehicles
- Modified driving style to accommodate identified vehicle problems and faults as described by the Assessor
- Followed emergency procedures for a situation described by the Assessor.

Summary

Drive vehicles

When driving vehicles:

- Practice in the vehicle to be used
- Adjust driving seat to individual need
- Adjust mirrors
- Conduct pre-trip inspections
- Drive defensively
- Plan ahead: think first, then act
- Constantly monitor the road and other road users
- Always use signals/indicators
- Accelerate and brake smoothly
- Avoid constant lane changes
- Try to create a safe space around the vehicle
- Match driving techniques to road and traffic conditions
- Use the mirrors
- Remember the length and weight of the vehicle and these impact vehicle performance
- Take extra care when towing a trailer and/or reversing
- Follow organisational protocols and vehicle instructions when shutting down and parking
- Be alert to potential on-the-road hazards and respond accordingly
- Adjust driving style to accommodate identified vehicle faults but never jeopardise passenger safety
- Follow emergency procedures as prescribed by the Tour Operator.

Presentation of written work

1. Introduction

It is important for students to present carefully prepared written work. Written presentation in industry must be professional in appearance and accurate in content. If students develop good writing skills whilst studying, they are able to easily transfer those skills to the workplace.

2. Style

Students should write in a style that is simple and concise. Short sentences and paragraphs are easier to read and understand. It helps to write a plan and at least one draft of the written work so that the final product will be well organised. The points presented will then follow a logical sequence and be relevant. Students should frequently refer to the question asked, to keep 'on track'. Teachers recognise and are critical of work that does not answer the question, or is 'padded' with irrelevant material. In summary, remember to:

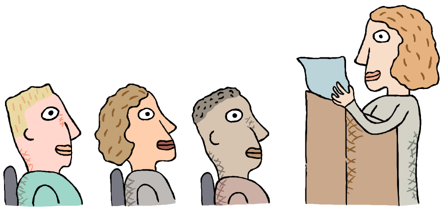
- Plan ahead
- Be clear and concise
- Answer the question
- Proofread the final draft.

3. Presenting Written Work

Types of written work

Students may be asked to write:

- Short and long reports
- Essays
- Records of interviews
- Questionnaires
- Business letters
- Resumes.



Format

All written work should be presented on A4 paper, single-sided with a left-hand margin. If work is word-processed, one-and-a-half or double spacing should be used. Handwritten work must be legible and should also be well spaced to allow for ease of reading. New paragraphs should not be indented but should be separated by a space. Pages must be numbered. If headings are also to be numbered, students should use a logical and sequential system of numbering.

Cover Sheet

All written work should be submitted with a cover sheet stapled to the front that contains:

- The student's name and student number
- The name of the class/unit
- The due date of the work
- The title of the work
- The teacher's name
- A signed declaration that the work does not involve plagiarism.

Keeping a Copy

Students must keep a copy of the written work in case it is lost. This rarely happens but it can be disastrous if a copy has not been kept.

Inclusive language

This means language that includes every section of the population. For instance, if a student were to write 'A nurse is responsible for the patients in her care at all times' it would be implying that all nurses are female and would be excluding male nurses.

Examples of appropriate language are shown on the right:

Mankind	*Humankind*
Barman/maid	*Bar attendant*
Host/hostess	*Host*
Waiter/waitress	*Waiter or waiting staff*

Recommended reading

Collins, Ben, 2014, How to drive [e-book], ISBN: 978174353233

Cressman, Hunter, Iacono, Paul, 2007, Defensive driving [DVD]

Denton, Tom, 2011, Automobile mechanical and electrical systems, ISBN: 9780080969459

Healey, Justin, 2009, Safe driving, ISBN: 9781921507083

Lyon, John. 2012, Advanced driving how to further skill and enjoyment in motoring, ISBN: 9780857332219

May, Ed, 2010, Automotive mechanics, ISBN: 9780070271876 9780070271883

O'Sullivan, Kerry, 20123, Learning to drive the LTRENT way, ISBN: 9780646573267

Victoria. Transport Accident Commission, 2004, Drive smart [CD-ROM]: improve your driving skills before you take the test, Version 2.0

Withers, Paula, Hutchinson, Tim, 2011, Safe driving, TAFE Queensland

Trainee evaluation sheet

Drive various types of service vehicles

The following statements are about the competency you have just completed.

Please tick the appropriate box	Agree	Don't Know	Do Not Agree	Does Not Apply
There was too much in this competency to cover without rushing.	☐	☐	☐	☐
Most of the competency seemed relevant to me.	☐	☐	☐	☐
The competency was at the right level for me.	☐	☐	☐	☐
I got enough help from my trainer.	☐	☐	☐	☐
The amount of activities was sufficient.	☐	☐	☐	☐
The competency allowed me to use my own initiative.	☐	☐	☐	☐
My training was well-organised.	☐	☐	☐	☐
My trainer had time to answer my questions.	☐	☐	☐	☐
I understood how I was going to be assessed.	☐	☐	☐	☐
I was given enough time to practice.	☐	☐	☐	☐
My trainer feedback was useful.	☐	☐	☐	☐
Enough equipment was available and it worked well.	☐	☐	☐	☐
The activities were too hard for me.	☐	☐	☐	☐

The best things about this unit were:

The worst things about this unit were:

The things you should change in this unit are:

Trainee self-assessment checklist

As an indicator to your Trainer/Assessor of your readiness for assessment in this unit please complete the following and hand to your Trainer/Assessor.

Drive various types of service vehicles

		Yes	No*
Element 1: Obtain driver’s licence(s)			
1.1	Identify the vehicles that have to be driven	☐	☐
1.2	Identify the driver’s licence(s) that need to be obtained	☐	☐
1.3	Undertake training to obtain the necessary licence(s)	☐	☐
1.4	Undertake driver licence assessment successfully	☐	☐
Element 2: Monitor service vehicles			
2.1	Perform regular preventative maintenance and operational checks of designated small vehicles	☐	☐
2.2	Complete and retain service reports as required	☐	☐
2.3	Adhere to scheduled maintenance and service requirements as set out by the vehicle manufacturer	☐	☐
2.4	Conduct special checks and service after designated trips	☐	☐
2.5	Check performance and efficiency of vehicle during operation	☐	☐
2.6	Undertake minor routine repairs	☐	☐
Element 3: Monitor road conditions			
3.1	Assess road conditions	☐	☐
3.2	Assess traffic conditions	☐	☐
3.3	Feedback information to head office about road and traffic conditions	☐	☐
3.4	Identify factors that may cause delays or route deviations	☐	☐
3.5	Factor conditions into choice of route to be taken	☐	☐
3.6	Read a map	☐	☐

		Yes	No*
Element 4: Drive vehicles			
4.1	Demonstrate basic operational driving practices	☐	☐
4.2	Demonstrate practices related to the towing of a trailer	☐	☐
4.3	Demonstrate reversing practices	☐	☐
4.4	Demonstrate parking and shutting down procedures	☐	☐
4.5	Identify and modify driving techniques to accommodate driving hazards	☐	☐
4.6	Modify driving style to accommodate identified vehicle problems and faults	☐	☐
4.7	Follow emergency procedures	☐	☐

Statement by Trainee:

I believe I am ready to be assessed on the following as indicated above:

Signed: ______________________________ **Date:** ____________

Note:

For all boxes where a **No*** is ticked, please provide details of the extra steps or work you need to do to become ready for assessment.

William Angliss Institute

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